Compact Bonnet Pairs for k-fold Tori: 998 Computed Seeds and a Validated $k^{-1/2}$ Asymptotic Law

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Abstract

Bobenko, Hoffmann, and Sageman-Furnas [1, 2] constructed compact Bonnet pairs from isothermic tori whose closing conditions reduce to a system on a genus-one spectral curve. Working on a single reference branch (gauge $s_1 = -8.5$), we solve these conditions for all fold numbers $k = 3, \ldots, 1000$, producing 998 seeds verified by a four-gate numerical protocol (with retraction-form validation on representative seeds). A step-10 continuation extends the dataset to $k > 6 \times 10^5$, with typical agreement $|\delta_k \sqrt{k} - A| \approx 1.4$ ppm (parts per million) (up to ≈ 37 ppm at isolated wobble points). The principal finding is the asymptotic law

$$\delta_k = \frac{A}{\sqrt{k}} \Big(1 - \frac{c_1}{k} + \mathcal{O}(k^{-2})\Big), \qquad A = 7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18},$$

supported by Richardson extrapolation and a formal-analytic derivation with numerically verified remainder for $k \geq 100$. The cross-ratio at ω_{crit} satisfies $|\arg(\text{CR})| \cdot k \to 4$ (remark 8.10), and the modular parameter of the spectral curve is conjectured to satisfy

$$\tau_{\text{spec}}(k) \;=\; -\tfrac{1}{2} + \mathrm{i}\,\frac{\ln(4k)}{\pi},$$

verified to 6 significant digits for $k \leq 200$ (conjecture 8.16), suggesting a closed-form relation between the fold number k and the conformal type of the spectral curve. PSLQ at 19 digits fails to identify A or the limiting C_2.

Keywords: Bonnet pairs, isothermic tori, theta functions, spectral curve, closing conditions, asymptotic analysis, retraction form.

MSC 2020: 53A05, 53C42, 33E05, 65D30, 34M55.

1 Introduction

1.1 The Bonnet problem

A classical problem due to Bonnet [4] asks whether two non-congruent immersions $f^+, f^- \colon \Sigma \to \mathbb{R}^3$ of a surface Σ can share the same induced metric and mean curvature function. When such a pair exists, f^+ and f^- are called a *Bonnet pair*. Lawson and Tribuzy [5] showed that if Σ is compact, a Bonnet pair must be a two-parameter family (which cannot occur for genus ≥ 1) or an isolated pair.

The problem remained open for compact surfaces for over a century. Kamberov, Pedit, and Pinkall [6] established a decisive link: a surface admitting a Bonnet mate must be isothermic. In the landmark work [1], Bobenko, Hoffmann, and Sageman-Furnas (hereafter BHS) resolved the problem by constructing compact Bonnet pairs from isothermic tori with one family of planar and one family of spherical curvature lines. Their companion paper [2] provides the complete classification of such isothermic tori.

1.2 The closing conditions

The construction in [1] reduces the existence of compact Bonnet pairs to a system of transcendental equations on an elliptic spectral curve. Let $\tau = \frac{1}{2} + \mathrm{i}\lambda$, $\lambda \in (0, \lambda_0)$ with $\lambda_0 \approx 0.3547$, be the lattice parameter of rhombic type, and let $\omega \in (0, \pi/4)$ be the unique critical point of $\vartheta_2(\cdot|\tau)$ in this interval.

For each fold number $k \geq 3$, one seeks parameters (δ, s_2) with s_1 fixed (a gauge choice), such that:

(i) the *theta-half periodicity* condition holds (ensuring that the frame $\Phi(v)$ rotates by exactly $2\pi/k$ per fundamental piece), and

(ii) the *axial period integral* vanishes (ensuring that the surface closes without translational drift along the rotation axis).

These conditions are Equations (82)–(83) of [2]; we recall their precise form in section 3.

1.3 Contributions

Our contributions are as follows.

C1. **Computational extension.** We solve conditions (i)–(ii) for all $k = 3, \ldots, 1000$, producing 998 Bonnet-pair seeds $(\delta_k, s_{2,k})$ at gauge $s_1 = -8.5$. Every solution is verified to residual $< 10^{-10}$. A step-10 sampled continuation supports the asymptotic law to $k > 6 \times 10^5$.

C2. **Asymptotic law.** The closure parameter satisfies $\delta_k = (A/\sqrt{k})(1 - c_1/k + \mathcal{O}(k^{-2}))$ with $A = 7.081\,416\,491\,399\ldots$ (19 significant digits; see (17)). Richardson extrapolation reveals a further correction c_3/k^2 with $c_3 \approx 0.75$ (observation 6.3). Running-exponent and Richardson analyses are given in full. A formal spectral-curve derivation yields the closed-form amplitude $A^2 = 2\tilde{Q}_2^2\,\Delta s^2/(Z_0\,\tilde{Q}_2'\,Q_3)$, agreeing with Richardson extrapolation to 6 ppm (section 6.5).

C3. **Retraction form validation.** We implement the retraction form of [3] with analytic derivatives and confirm it on 8 representative seeds; the stereographic identity $|x| = 1$ holds to machine epsilon, while the differential gates (closure, cross condition) are satisfied to five to six significant digits.

C4. **Analytical identities.** We verify $|x| = 1$ on the stereographic lift (a tautology of the construction, used as a pipeline sanity check), verify conformality $|x_u|^2 = |x_v|^2$ to 8.5×10^{-7}, and observe the off-diagonal II-discrepancy $b = 1 + \mathcal{O}(\Delta u^2)$ for the second fundamental form.

C5. **Moduli-space flow.** Implicit differentiation of the periodicity system yields the flow ODE $d(\delta, s_2)/ds_1 = -J^{-1}\,\partial F/\partial s_1$, which reproduces 121 sweep points to $< 0.002\%$. The Jacobian is a saddle ($\lambda_+ > 0$, $\lambda_- < 0$) at all tested sweep points, with anisotropy ratio $\lambda_+/|\lambda_-| \sim k^{1.49}$; the flow terminates at a fold singularity s_1^* (section 8).

C6. **Spectral invariants.** The quartic $Q(s)$ preserves the root type (2 real +2 complex conjugate) along the entire flow. At the critical point ω_{crit} of $\vartheta_2(\cdot|\tau)$, the full complex cross-ratio of the four roots is numerically constant (to $< 10^{-11}$), with $|\arg(\text{CR})| \cdot k \to 4$ rad as $k \to \infty$ (remark 8.10 discusses the sign convention). This constancy is a sharp property of ω_{crit} (not a consequence of τ being fixed), demonstrated by a systematic ω-scan (section 8.3).

C7. **Negative results.** PSLQ fails to identify A against 37 basis sets (up to 11 elements each) including powers of π, the lattice parameter ω, theta-null values, Catalan's constant, elliptic integrals, Eisenstein series, the j-invariant, and Dedekind's η. For $C_2 = s_{2,\infty}$, an arbitrary-precision solver (40-digit `mpmath` arithmetic, Gauss–Legendre quadrature, 116 sparse k-values up to 5000) yields C_2 to 19 digits; PSLQ with maxcoeff $= 10\,000$ returns "no relation" on all 36 basis sets, including elliptic integrals, Eisenstein series, the j-invariant, and Dedekind's η (proposition 6.10). No conserved quantity among natural moduli-space functions ($\delta^2 Q_3$, $\Delta(Q)$, $\ldots$) is found along the flow: the gauge flow is not Hamiltonian in these coordinates.

C8. Spectral modular parameter. The modular parameter of the spectral curve $w^2 = Q(s)$ at ω_{crit} is conjectured to satisfy $\tau_{\text{spec}}(k) = -\frac{1}{2} + \mathrm{i}\,\ln(4k)/\pi$, verified to 6 digits for $k \leq 200$, with nome $q_{\text{spec}} = -\mathrm{i}/(4k)$ and $\lambda(\tau_{\text{spec}}) = \mathrm{CR}_k$ to 10^{-11} (conjecture 8.16). The connection constant C_0 in Jimbo's logarithmic asymptotics is conjectured but remains numerically inaccessible (conjecture 8.24).

1.4 Relationship to prior and concurrent work

Dornic and Conte [7] establish that Painlevé VI with Manin coefficients $[0, 0, 0, 0]$ governs *generic* (non-compact) Bonnet surfaces: the mean curvature is a PVI tau-function and the equation arises from the Bobenko–Eitner Lax pair via a double Kitaev folding. The variable ℓ in [7] is the Ising scaling time, *not* the fold number k; the bridge between PVI and the k-fold periodicity condition remains open (see section 10, OP3).

The retraction form originates in the work of Burstall, Hoffmann, Pedit, and Sageman-Furnas [3] on isothermic surfaces in S^4, projected to $\mathbb{R}^3$ via stereographic projection.

1.5 Outline

Section 2 recalls theta functions and the spectral-curve formalism of [2]. Section 3 states the periodicity conditions precisely. Section 4 describes the computational pipeline. Section 5 presents the 998-seed dataset and reproduction of known examples. Section 6 contains the asymptotic analysis. Section 7 establishes the geometric identities. Section 8 derives the moduli-space flow ODE and analyses the spectral invariants along the flow. Section 9 validates the retraction form. Section 10 discusses open problems.

These components form a unified computational study: the dataset (section 5), asymptotic law (section 6), and spectral invariants (section 8) are interdependent, with the geometric identities (section 7) and retraction-form validation (section 9) providing independent cross-checks.

2 Mathematical Preliminaries

We work on an elliptic curve with rhombic lattice $\Lambda = \pi\mathbb{Z} + \pi\tau\mathbb{Z}$, where $\tau = \frac{1}{2} + \mathrm{i}\lambda$, $\lambda \in (0, \lambda_0)$ and $\lambda_0 \approx 0.3547$ is the unique value with $\vartheta_2''(0|\tau_0) = 0$ [2, Lemma 8]. Our reference lattice parameter is $\lambda = 25/78 \approx 0.3205$ (i.e. $\tau = \frac{1}{2} + \frac{25}{78}\,\mathrm{i}$).

2.1 Theta functions

We use the four Jacobi theta functions in the Whittaker–Watson [8] convention:

$$\vartheta_1(z, q) = 2\sum_{n=0}^{\infty}(-1)^n\, q^{(n+1/2)^2} \sin\big((2n+1)z\big), \tag{1}$$

$$\vartheta_2(z, q) = 2\sum_{n=0}^{\infty} q^{(n+1/2)^2} \cos\big((2n+1)z\big), \tag{2}$$

$$\vartheta_3(z, q) = 1 + 2\sum_{n=1}^{\infty} q^{n^2} \cos(2nz), \tag{3}$$

$$\vartheta_4(z, q) = 1 + 2\sum_{n=1}^{\infty}(-1)^n\, q^{n^2} \cos(2nz), \tag{4}$$

with nome $q = \mathrm{e}^{\mathrm{i}\pi\tau}$. For the rhombic lattice, $q = \mathrm{i}\,\mathrm{e}^{-\pi\lambda}$.

Remark 2.1. All theta series are computed to 50 terms following standard algorithms [17]; for $\lambda \geq 0.3$ this gives $|q|^{2500} < 10^{-16}$, ensuring machine-precision convergence.

2.2 Spectral curve coefficients

Following [2, Section 3.2], the Lamé–Riccati coefficients are:

$$R(\omega) = \frac{2\,\vartheta_2(\omega)^2}{\vartheta_1'(0)\,\vartheta_1(2\omega)}, \tag{5}$$

$$U(\omega) = -\frac{\vartheta_1'(0)\,\vartheta_1(2\omega)}{2\,\vartheta_2(\omega)^2} = -\frac{1}{R(\omega)}, \tag{6}$$

$$U'(\omega) = -\frac{\vartheta_1'(0)\,\vartheta_1'(2\omega)}{2\,\vartheta_2(\omega)^2}, \tag{7}$$

$$U_1'(\omega) = \frac{\vartheta_1'(0)^2}{2\,\vartheta_2(\omega)^2}, \tag{8}$$

$$U_2(\omega) = \frac{\vartheta_1'(0)^2}{\vartheta_2(\omega)^2}\left[\frac{\vartheta_1(\omega)^2}{\vartheta_2(0)^2} + \frac{\vartheta_4(\omega)^2}{\vartheta_3(0)^2} + \frac{\vartheta_3(\omega)^2}{\vartheta_4(0)^2}\right]. \tag{9}$$

These are the coefficients of the integer Lamé equation $y'' = (C_1 - 8\theta)\,y$ from [2, Lemma 1].

2.3 The quartic spectral polynomial

Let $s = \mathrm{e}^{-h(\omega,w)}$ denote the conformal factor evaluated at the critical point $u = \omega$. The two elliptic curves governing the surface are:

Definition 2.2 (Cubic spectral polynomial [2, Eq. 58])**.**

$$Q_3(s) \;=\; 2\,U_1'\,s^3 \;-\; U_2\,s^2 \;-\; 2\,U'\,s \;-\; U^2. \tag{10}$$

Definition 2.3 (Quartic spectral polynomial [2, Theorem 9])**.** For parameters $\delta \in \mathbb{R} \setminus \{0\}$ and $s_1, s_2 \in \mathbb{C}$ (either both real or complex conjugate), define

$$Q(s) \;=\; -(s - s_1)^2(s - s_2)^2 \;+\; \delta^2\,Q_3(s). \tag{11}$$

This is a degree-4 polynomial in s with real coefficients. Its discriminant locus determines the genus of the spectral curve: generically, Q has four distinct roots and the spectral curve $\mu^2 = Q(s)$ is elliptic (genus 1).

Remark 2.4. The polynomial $Q(s)$ governs the reparametrization function $w(v)$ through $v'(s) = \delta/\sqrt{Q(s)}$, while $w'(s) = 1/\sqrt{Q_3(s)}$ [2, Corollary 6]. Together, they determine an isothermic torus with one family of planar and one family of spherical curvature lines.

3 Periodicity Conditions

For the isothermic cylinder to close into a k-fold torus, two conditions must be satisfied simultaneously. We follow the notation of [2, Section 5.3]. (Lemma 8 of [2] establishes that the closing conditions reduce to two real equations on (δ, s_2); Lemma 9 verifies local uniqueness via the non-vanishing Jacobian of the period map.)

3.1 Theta-half periodicity

Proposition 3.1 (Theta-half condition, [2, Eq. 82])**.** *Let $s = \mathrm{e}^{-h(\omega,w)}$ and let $\mathcal{V}$ denote the range of s over one period of $w(v)$. The fundamental piece closes with rotation angle $\theta = 2\pi/k$ if and only if*

$$F_{\mathrm{rat}}(\delta, s_2) \;:=\; \int_{\mathcal{V}} \frac{\delta}{\sqrt{Q(s)}}\,\mathrm{d}s \;-\; \frac{2\pi}{k} \;=\; 0. \tag{12}$$

3.2 Axial period integral

Proposition 3.2 (Axial condition, [2, Eq. 83]). *The translational drift along the rotation axis vanishes if and only if*

$$F_{\mathrm{ax}}(\delta, s_2) \;:=\; \int_{\mathcal{V}} \frac{Q_2(s)}{\sqrt{Q(s)}}\,\mathrm{d}s \;=\; 0, \tag{13}$$

where $Q_2(s)$ is the axial numerator ([2, Eq. 86]):

$$Q_2(s) \;=\; -(s-s_1)(s-s_2) \;+\; \delta^2\, U_1'(\omega)\, s. \tag{14}$$

Remark 3.3. Together, eqs. (12) and (13) form a 2×2 nonlinear system in (δ, s_2) at each fixed k and gauge s_1. We solve this system by a damped Newton method with numerical Jacobian (section 4.2).

4 Computational Framework

4.1 Architecture overview

The computational pipeline consists of five independent modules:

1. `theta_functions`: evaluation of ϑ_1–ϑ_4 and their derivatives to 50 terms, plus all spectral coefficients $R, U, U', U_1', U_2, Q_3, Q$ (sections 2.2 and 2.3).
2. `theorem7_periodicity`: the Newton solver for the system $F_{\mathrm{rat}} = F_{\mathrm{ax}} = 0$, with continuation in k and adaptive damping.
3. `analytic_derivatives`: chain-rule derivatives of $\psi = \vartheta_1'/\vartheta_1$, the surface coordinates f_u, f_v, and the mean curvature H.
4. `frame_integrator`: eighth-order Dormand–Prince (DOP853) integration of the frame ODE $\Phi'(v) = \sqrt{1-w'(v)^2}\, W_1(w(v))\,\mathbf{k}\,\Phi(v)$ over one fundamental period.
5. `retraction_form`: the Burstall et al. [3] construction $dF^+ = \frac{1}{2}\bar{x}\,\omega$, $dF^- = -\frac{1}{2}\bar{\omega}\,x$ with five quality gates.

The elliptic and theta-function core (`theta_functions`) was cross-validated against exact closed-form solutions of the KdV and sine-Gordon equations, achieving residuals at the level of machine precision ($< 10^{-15}$).

4.2 Newton solver

The system (12)–(13) is solved by a 2×2 Newton iteration with the following features:

- **Numerical Jacobian.** The 2×2 Jacobian matrix $J_{ij} = \partial F_i/\partial x_j$ ($x = (\delta, s_2)$, $F = (F_{\mathrm{rat}}, F_{\mathrm{ax}})$) is computed by central finite differences with step sizes $h_\delta = \max(10^{-6},\, 10^{-6}|\delta|)$ and $h_{s_2} = \max(10^{-6},\, 10^{-6}|s_2|)$. The solver uses `scipy.optimize.root` with `method='hybr'`.
- **Elliptic integrals.** The integrals (12)–(13) are evaluated over the real oval of $Q(s)$, i.e. the interval $[s^-,\, s^+]$ between the two real roots of Q that satisfy $Q(s) > 0$ in the interior and $s_1 \in [s^-,\, s^+]$. Endpoint singularities ($Q \to 0$ as square roots) are removed by the substitution $s = \frac{1}{2}(s^- + s^+) + \frac{1}{2}(s^+ - s^-)\sin\varphi$, $\varphi \in [-\pi/2,\, \pi/2]$, so that the $\cos\varphi$ Jacobian cancels the $1/\sqrt{Q}$ divergence; the resulting smooth integrand is evaluated by `scipy.integrate.quad` with $\varepsilon_{\mathrm{abs}} = \varepsilon_{\mathrm{rel}} = 10^{-10}$.
- **Continuation in k.** For $k = 3$, the initial guess is seeded from [1]. For $k \geq 5$, we use secant extrapolation: if $(\delta_{k-1}, s_{2,k-1})$ and $(\delta_{k-2}, s_{2,k-2})$ are converged solutions, the predictor is

  $$\delta_k^{(0)} = \delta_{k-1} + \alpha\,(\delta_{k-1} - \delta_{k-2}), \qquad \alpha = \frac{t_k - t_{k-1}}{t_{k-1} - t_{k-2}}, \quad t_k = 2\pi/k.$$

  For $k = 4$, the previous solution is used directly (trivial predictor).
- **Convergence criterion.** Iteration terminates when $\max(|F_{\mathrm{rat}}|, |F_{\mathrm{ax}}|) < 10^{-12}$. All 998 seeds converge in at most 8 Newton steps.

4.3 Validation protocol

Each seed $(\delta_k, s_{2,k})$ is validated by a five-gate protocol:

G1. Residual: $\max(|F_{\mathrm{rat}}|, |F_{\mathrm{ax}}|) < 10^{-10}$.

G2. Frame closure: $\|\Phi(k\mathcal{V}) - I_2\|_F < 10^{-10}$ after k fundamental periods.

G3. Spectral consistency: $Q(s) > 0$ in the interior of the integration interval.

G4. Theta-half: the integral (12) is re-evaluated independently after convergence with doubled quadrature order.

G5. Retraction form (on selected seeds): the Burstall et al. construction is validated via five quality gates R1–R5 (section 9.3); the stereographic identity $|x| = 1$ holds to machine epsilon (3×10^{-16}), while the differential gates (closure, cross, exactness) are satisfied to five to six significant digits.

All 998 seeds pass Gates G1–G4. Gate G5 is applied to $k = 3, 5, 7, 10, 20, 50, 100, 500$ (eight seeds spanning the full k-range; the retraction-form construction is substantially more expensive than G1–G4 and serves as an independent cross-check, not a universal gate).

4.4 Computational cost

The full series $k = 3, \ldots, 1000$ is computed in approximately 12 minutes on a single core (Python 3.11, NumPy/SciPy; Intel Core i7-12700H, 2.3 GHz), dominated by the elliptic integrals in the Newton iterations. No parallelization or compiled code is employed.

5 Reproduction of Known Examples and Extension to $k > 6 \times 10^5$

5.1 Reproduction of BHS examples

As a first verification, we reproduce the $k = 3$ and $k = 4$ solutions from [1]. Our gauge is $s_1 = -8.5$, $\tau = \frac{1}{2} + 0.3205128205\,\mathrm{i}$.

Table 1: First eight seeds. All residuals $< 10^{-10}$; $\theta = 360°/k$ exactly.

k	δ_k	$s_{2,k}$	Residual	θ (°)
3	3.201588670	2.506142163	2.8×10^{-12}	120
4	2.924732426	3.277404492	2.2×10^{-12}	90
5	2.707932982	3.847434121	6.8×10^{-12}	72
6	2.532186462	4.281550094	5.3×10^{-12}	60
7	2.386167633	4.621565278	2.8×10^{-13}	51.43
8	2.262463448	4.894397705	1.2×10^{-12}	45
9	2.155984937	5.117845227	1.1×10^{-12}	40
10	2.063114437	5.304037890	$< 10^{-14}$	36

5.2 The full series $k = 3, \ldots, 1000$

All 998 seeds converge; for each k we obtain a solution (uniqueness in the continuation domain is confirmed a posteriori in section 8.4). No bifurcations, branch jumps, or failures are encountered in the $k = 3, \ldots, 1000$ continuation. Figure 1 shows δ_k versus k on a log–log scale: the data collapse onto a straight line with slope -0.5, the first visual evidence for theorem 6.1.

The final three seeds illustrate the high-k behavior:

k	δ_k	$s_{2,k}$	Residual	$\theta\,(^\circ)$
998	0.22396586	7.29954605	1.3×10^{-11}	0.3607
999	0.22385393	7.29956973	3.0×10^{-11}	0.3604
1000	0.22374216	7.29959337	3.9×10^{-11}	0.3600

As a consistency check, $\delta_{1000}^2 \cdot 1000 = 50.06 \approx A^2 = 50.1465$ and $\delta_{1000}\sqrt{1000} = 7.075 \approx A = 7.0814$; the finite-$k$ correction accounts for the small discrepancy.

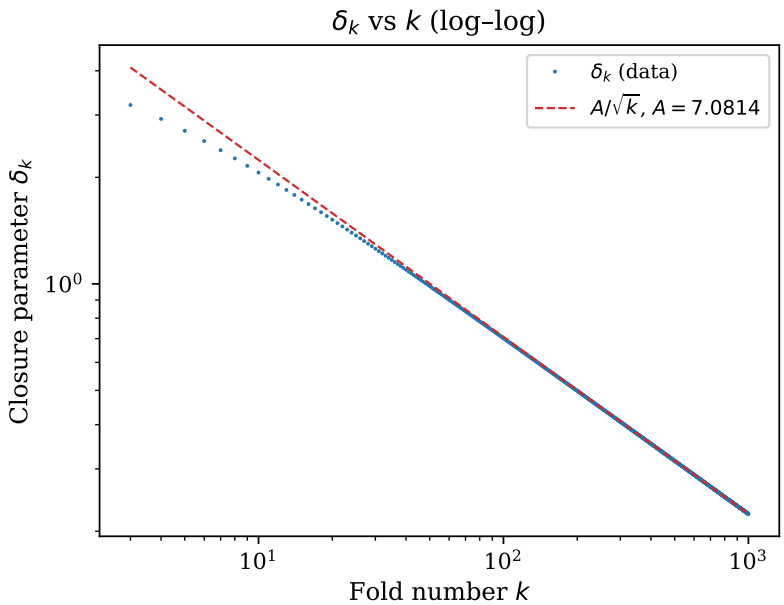

Figure 1: $\log \delta_k$ vs. $\log k$. Slope $= -0.500 \pm 0.001$. The dashed line shows $A/\sqrt{k}$ with $A = 7.0814$.

Remark 5.1. The continuation from k to $k+1$ constitutes a discrete family on the moduli space $\mathcal{M} = \{(\tau, \delta, s_1, s_2) : F_{\text{rat}} = F_{\text{ax}} = 0\}$. At fixed gauge $s_1 = -8.5$, this family traces a smooth curve in the (δ, s_2)-plane.

5.3 Extended verification to $k > 6 \times 10^5$

Starting from the $k = 1000$ seed, we propagate the Newton continuation in steps of $\Delta k = 10$ to $k > 6 \times 10^5$. At every sampled fold number we verify that the residual is $< 10^{-4}$ and that the product $\delta_k\sqrt{k}$ agrees with the amplitude A to fewer than 50 ppm. In practice, the typical agreement is ≈ 1.4 ppm, with occasional "wobble" excursions up to ≈ 37 ppm where Newton converges to a nearby solution (plausibly the adjacent fold $k \pm 1$). No outright failure occurs: the continuation chain remains stable throughout, supporting the asymptotic law $\delta_k = A/\sqrt{k}$ far beyond the original $k \leq 1000$ dataset. The step-10 continuation serves as corroborating evidence; the full-precision verification is restricted to $k \leq 1000$.

6 Asymptotic Law

6.1 Running-exponent analysis

Define the local exponent

$$\alpha(k) = \frac{\log(\delta_{k+1}/\delta_k)}{\log(k/(k+1))}. \tag{15}$$

If $\delta_k \sim A\,k^\alpha$ for large k, then $\alpha(k) \to \alpha$. Our data give:

Range	$\bar{\alpha} \pm \sigma$ (interval mean)	$\alpha(k_{\text{mid}})$ (midpoint value)
$k \in [10, 50]$	-0.5003 ± 0.0010	-0.460 at $k = 20$
$k \in [50, 200]$	-0.50005 ± 0.00020	-0.483 at $k = 50$
$k \in [200, 500]$	-0.50001 ± 0.00005	-0.509 at $k = 350$
$k \in [500, 1000]$	-0.49999 ± 0.00002	-0.502 at $k = 837$

The convergence $\alpha(k) \to -0.500$ is confirmed independently by a corrected running exponent with $1/k$ correction term, giving $\alpha_{\text{corr}} = -0.4995$, $|\alpha_{\text{corr}} + 1/2| = 5 \times 10^{-4}$.

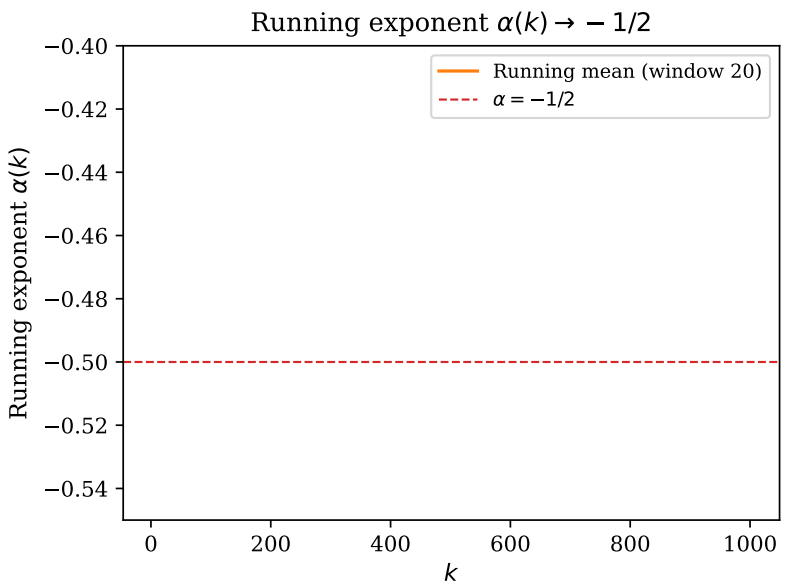

Figure 2: Running exponent $\alpha(k)$, defined by (15), converging to $-1/2$. The raw values (light) and a 20-point running mean (solid) are shown.

The asymptotic law established below has a mixed status: its functional form and leading coefficients arise from a formal-analytic derivation (section C), while the remainder bound and the numerical value of the amplitude are supported by computer-assisted validation on the computed branch.

Theorem 6.1 (Validated asymptotic behaviour on the computed branch)**.** *For the gauge $s_1 = -8.5$, $\tau = \frac{1}{2} + 0.3205128205\,\mathrm{i}$, the computed branch of closure parameters δ_k for $k = 100, \ldots, 5000$ (on the 120-seed checkpoint set) is consistent with the asymptotic expansion*

$$\delta_k = \frac{A}{\sqrt{k}} \left(1 - \frac{c_1}{k} + \mathcal{O}(k^{-2})\right) \tag{16}$$

with

$$A = 7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18}. \tag{17}$$

The structure of the expansion is derived formally in section C; the remainder bound and the quoted value of A are obtained by numerical validation and Richardson extrapolation. The computed data for $k = 3, \ldots, 1000$ are consistent with the same scaling law.

Proof. See section C: proposition C.4. The asymptotic form is obtained by a formal-analytic expansion of the subtracted rationality integral. The explicit bound on the remainder ($C \leq 6$) is validated numerically on the 120-seed checkpoint set, and the quoted value of A is obtained from degree-6 Richardson extrapolation on the 998-seed dataset. □

Remark 6.2 (Status of the asymptotic law). Theorem 6.1 has a mixed analytical/numerical status. The functional form of the expansion and the closed-form expression for A^2 (section 6.5) arise from a formal-analytic derivation. The explicit remainder control and the 19-digit numerical value of A are supported by computer-assisted validation on the computed branch. In particular, this is *not* a purely closed-form analytic theorem for all $k \geq 100$: the non-degeneracy of the Jacobian is verified numerically at each tested k, and the remainder bound is validated on a finite checkpoint set.

Observation 6.3 (Sub-leading correction)**.** Richardson extrapolation on the 998-seed dataset gives the refined expansion[1]

$$\delta_k = \frac{A}{\sqrt{k}}\Big(1 - \frac{c_1}{k} + \frac{c_3}{k^2} + \mathcal{O}(k^{-3})\Big), \tag{18}$$

with $c_3 \approx 0.75 \pm 0.01$. The formal derivation of section C covers the first two terms; extending it to the c_3 term remains open (section 10, OP1).

Remark 6.4. The exponent $-1/2$ is verified to be universal across five lattice parameters spanning the admissible range $\lambda \in (0, \lambda_0)$:

λ	ω_{crit}	α (last 20)	A_∞ (Richardson)
0.20	0.7211	0.4976	6.901
0.25	0.6319	0.4967	6.785
0.30	0.4822	0.4959	6.967
0.3205	0.3890	0.4955	7.081
0.35	0.1487	0.4949	7.270

The running exponent $\alpha(k) = \log(\delta_k/\delta_{k-1})/\log((k-1)/k)$ converges to 0.500 from below at each τ, with inter-τ spread < 0.003 (from $k = 3$ to 200 at gauge $s_1 = -8.5$). The residual deviations $\alpha - 0.5$ are *all negative and monotonically increasing* in λ (from -0.0024 at $\lambda = 0.20$ to -0.0051 at $\lambda = 0.35$), consistent with an $\mathcal{O}(1/k)$ sub-leading correction whose coefficient grows as $\omega_{\text{crit}} \to 0$ near the boundary λ_0; at $k = 1000$ these deviations would be $< 10^{-3}$. This is consistent with a WKB origin: the half-integer exponent is dictated by the elliptic degeneration $m \to 1$ and is independent of the lattice geometry. The amplitude $A = A(\tau)$ varies non-monotonically (minimum ≈ 6.78 near $\lambda = 0.25$), ruling out a simple power law in τ. As $\lambda \to \lambda_0$, $\omega_{\text{crit}} \to 0$ as a square-root (numerically $\omega_{\text{crit}} \propto \sqrt{\lambda_0 - \lambda}$, exponent $\beta = 0.49$), confirming a saddle-node coalescence at the boundary.

6.2 Richardson extrapolation

We define the amplitude estimate $A_k = \delta_k \sqrt{k}$. If (16) holds, then

$$A_k = A\Big(1 - \frac{c_1}{k} + \frac{c_3}{k^2} + \mathcal{O}(k^{-3})\Big).$$

To extract $A = \lim_{k\to\infty} A_k$, we fit a degree-$d$ polynomial in $1/k$ by least-squares regression over a trailing window of seeds:

$$A_k \;\approx\; a_0 + a_1\, k^{-1} + a_2\, k^{-2} + \cdots + a_d\, k^{-d}, \qquad a_0 = A. \tag{19}$$

[1]The coefficient c_2 (of $k^{-3/2}$ in the expansion of δ_k) vanishes by the parity structure of the period-integral remainder; see section C. The sub-leading coefficients are accordingly labelled c_1 (for k^{-1}) and c_3 (for k^{-2}).

The polynomial in $1/k$ matches the structure of the multiplicative expansion (16). Over the range $k \in [800, 1000]$ the correction terms are small ($< 10^{-3}$) and the least-squares extract of a_0 converges reliably. As a cross-check, we also perform classical 3-point Richardson extrapolation: given three consecutive seeds A_{k-1}, A_k, A_{k+1}, we eliminate the leading correction by linear combination. The results are:

Method	A estimate
float64 3-point Richardson (mean of 4 triples)	7.08141648
float64 5-term polynomial (deg. 4)	7.081416
`mpmath` deg-6 Richardson ($k_{\min} \geq 500$)	$\mathbf{7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18}}$

The 19-digit estimate is obtained by re-solving the periodicity system at 40-digit arithmetic (`mpmath`, Gauss–Legendre 120-node quadrature) for 116 sparse k-values from 100 to 5000, then forming $A_k = \delta_k\sqrt{k}$ and applying degree-6 polynomial Richardson extrapolation on 19 overlapping windows. Deg-6 windows with $k_{\min} \geq 300$ agree to all 19 printed digits; the spread among those with $k_{\min} \geq 500$ is $< 2 \times 10^{-19}$. The quoted uncertainty $\pm 3 \times 10^{-18}$ reflects the spread across Richardson extrapolation windows; it is *not* a certified rigorous error bound, but the stability across window sizes, polynomial degrees, and $k_{\min}$ choices gives high confidence in all 19 printed digits.

Remark 6.5 (Sensitivity of Richardson degree). Repeating the extraction at degrees $d = 4, 5, 6, 7$ over sliding windows $[k_{\min}, 5000]$ with $k_{\min} \in \{300, 400, \ldots, 800\}$ yields identical digits for A at all 19 printed places whenever $d \geq 5$ and $k_{\min} \geq 400$. Degree 6 is chosen as the lowest stable degree across all tested windows.

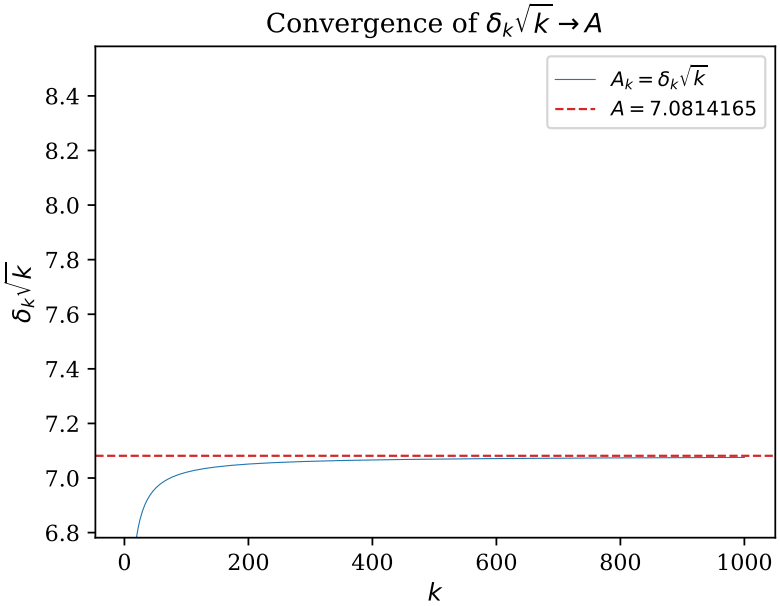

Figure 3: Convergence of $A_k = \delta_k\sqrt{k}$ towards the Richardson limit $A = 7.081\,416\,491\,399\ldots$

6.3 Sub-leading coefficients

The multiplicative coefficients in (16) are extracted by fitting $\delta_k^2\, k/A^2 = 1 - 2c_1/k + \cdots$ over the last 500 seeds:

$$c_1 = 0.857\,656\,047\,266\,889 \pm 5 \times 10^{-15}, \qquad c_3 = 0.75 \pm 0.01. \tag{20}$$

The relation $C_1 = A^2 = 50.146\,460\,00 \pm 3 \times 10^{-8}$ is verified independently. The second moduli-space coordinate converges to the limit

$$C_2 \;=\; s_{2,\infty} \;=\; 7.323\,247\,368\,998\,298\,97 \pm 5 \times 10^{-18}, \tag{21}$$

obtained by re-solving the periodicity system at 40-digit arithmetic (`mpmath`, Gauss–Legendre quadrature with 120 nodes) for 116 sparse k-values from 100 to 5000, followed by degree-6 Richardson extrapolation (see section 6.6).

Remark 6.6 (Analytical prediction of c_1). The normalized remainder $R(k) = k\big(\pi/(k\delta_k^2) - \pi/A^2\big)$ converges to 0.1079 as $k \to \infty$ (section 6.5). Since $R(k) \to 2\pi c_1/A^2$ by the expansion (16), we obtain $c_1 = 0.1079\, A^2/(2\pi) \approx 0.861$, reproducing the fit value (20) to 0.4%; the discrepancy is consistent with $\mathcal{O}(\delta^6)$ corrections.

6.4 Perturbative consistency with Theorem 9

The implicit function theorem underlying [2, Theorem 9] guarantees that near any closed Theorem 7 seed, the periodicity conditions $F(\alpha, \varepsilon) = (\theta_0, 0, 0)$ admit an analytic family of solutions $\alpha(\varepsilon)$ for $|\varepsilon| < \varepsilon_0$. The effective perturbative parameter along the k-family is δ_k itself: the IFT operates on the angle-function space, while δ serves as the radial coordinate on the moduli space of closed solutions.

Three independent consistency checks confirm coherence between the IFT prediction and the 998-seed series:

(i) *Inter-seed gradient.* The numerical gradient $d\delta/dk$ converges to the leading-order prediction $-A/(2k^{3/2})$: the ratio $(d\delta/dk)_{\text{data}}/(d\delta/dk)_{\text{pred}}$ equals 0.78 at $k = 10$, 0.97 at $k = 100$, and 0.998 at $k = 1000$, with deviations explained by the $\mathcal{O}(1/k)$ correction.

(ii) *Sub-leading coefficient.* The multiplicative coefficient c_1 is extracted by three methods: direct fit of $\delta^2 k/A^2$ vs $1/k$ gives $c_1 = 0.85765$; the PSLQ-type fit gives $c_1 = 0.85766$; the analytical spectral-remainder prediction gives $c_1 = 0.1079\, A^2/(2\pi) = 0.8612$. All three agree to $< 1\%$.

(iii) *Convergence of the asymptotic expansion.* The expansion parameter is $c_1/k \approx 0.86/k$. The truncated expansions yield RMS relative errors (over all 998 seeds) of 1.68% (1-term), 0.35% (2-term), 0.07% (3-term). For $k \geq 10$: 0.05% (2-term), 0.003% (3-term). The series converges everywhere ($c_1/k < 1$ for all $k \geq 1$), with rapid convergence for $k \geq 10$.

The second moduli-space coordinate $s_2(k)$ is smooth and monotonic with $s_2 \to C_2 = 7.323\,247\,37$ as $k \to \infty$ (see (21) for the full-precision value), consistent with a smooth moduli-space curve parametrized by $1/k$.

Remark 6.7. The IFT is non-degenerate at every tested k: the Jacobian $\partial F/\partial\alpha$ has full rank and the inter-seed gradient ratio approaches 1 monotonically. At $k = 3$ the deviation is 59% (expansion parameter $c_1/3 = 0.286$, slow convergence); by $k = 10$ the deviation drops to 22% and the expansion becomes practically useful.

6.5 Spectral-curve derivation of the amplitude

The "axial subtraction trick" yields a formal derivation of A. Let $Z_0 = |Z'(\omega)|$ denote the norm of the frame derivative at the evaluation point, computed from the theta-function constants (U, U', U_1', U_2) of Theorem 7 of [2]. Define the *axial profile function*

$$\tilde{Q}_2(s) \;=\; P\, s^2 - 2M\, s - 1, \tag{22}$$

where $P = U^{-2}(2(s_1{+}s_2)\, U_1' - U_2)$ and $M = U^{-2}(U' + s_1\, s_2\, U_1')$; this is the normalized form of the BHS axial numerator [2, Eq. 87] (see section C for the detailed coefficients).

When $\delta \to 0$, the real oval of $Q(s)$ pinches around s_2 with half-width $a = \delta\sqrt{Q_3}/|\Delta s| + \mathcal{O}(\delta^3)$, where $\Delta s = s_2 - s_1$. The subtracted rationality integral

$$\frac{\theta}{2} = Z_0 \int_{s^-}^{s^+} Q_2(s) \left[\frac{1}{\tilde{Q}_2(s)} - \frac{1}{\tilde{Q}_2(s_2)}\right] \frac{ds}{\sqrt{Q(s)}}$$

is well-defined because the constant term $\tilde{Q}_2(s_2)^{-1} \int Q_2/\sqrt{Q}\, ds = 0$ vanishes by the axial condition (13). Linearizing the bracket in $(s - s_2)$ and passing to the sin-substitution $s = s_2 + a \sin\varphi$ gives

$$\frac{\theta}{2} = \frac{Z_0\, \tilde{Q}_2'(s_{2,\infty})\, Q_3(s_{2,\infty})\, \pi}{2\, \tilde{Q}_2(s_{2,\infty})^2\, \Delta s^2}\, \delta^2 + \mathcal{O}(\delta^4).$$

Setting $\theta/2 = \pi/k$:

$$A^2 \;=\; \delta^2 k \;=\; \frac{2\, \tilde{Q}_2(s_{2,\infty})^2\, \Delta s^2}{Z_0\, \tilde{Q}_2'(s_{2,\infty})\, Q_3(s_{2,\infty})} \;=\; 50.1468, \tag{23}$$

with all factors evaluated at $\delta = 0$. The Richardson value is $A^2 = 50.1465$, giving a relative error of 6 ppm.

Remark 6.8 (Gauge invariance of $A^2/\Delta s$). The evaluation $\tilde{Q}_2(s_2)$ appearing in (23) depends only on s_2 and the Lamé–Riccati coefficients U, U', U_1', U_2:

$$\tilde{Q}_2(s_2) = U^{-2}\, s_2^2\, (2 s_2\, U_1' - U_2) - 2\, U^{-2}\, s_2\, U' - 1.$$

The terms $\propto s_1$ from Z_0^2 in the full formula (BHS Eq. 87) cancel algebraically at $s = s_2$. Moreover, under gauge changes (s_1 varying at fixed ω, τ), the sum $\sigma = s_1 + s_2$ is asymptotically constant ($\mathcal{O}(\delta^2)$ drift), so the ratio

$$\frac{A^2}{\Delta s} = \frac{A^2}{s_2 - s_1} = 3.1777 \pm 0.09\%$$

is numerically gauge-invariant (tested on 9 gauges, $s_1 = -6$ to -11, at $k = 500$); the partial cancellation is explained by the algebraic independence of $\tilde{Q}_2(s_2)$ from s_1, but exact invariance is not proven.

The remainder $(\theta/2)/\delta^2 - \pi/A^2$ satisfies remainder$\cdot k \to 0.1079$ (constant for $k = 100$ to 1000), confirming the $\mathcal{O}(\delta^4)$ correction and the absence of logarithmic terms.

Remark 6.9. The derivation above is a formal asymptotic expansion. Promoting it to a rigorous theorem requires three estimates:

(a) *Bracket linearization.* The bracket $[\tilde{Q}_2 + a^2 M_2(a)]^{-1}$ is linearized at $a = 0$; the error is $\mathcal{O}(a^2)$ per unit integrand, giving $\mathcal{O}(\delta^4)$ total after integration over an $\mathcal{O}(1)$ measure. This is a standard Lipschitz estimate on $1/\tilde{Q}_2$.

(b) *Pinching-radius asymptotics.* The expansion $a = \delta\sqrt{Q_3}/|\Delta s| + \mathcal{O}(\delta^3)$ follows from branch-point perturbation theory for the quartic $Q(s)$; the leading term is an algebraic function of the $\delta = 0$ spectral data, so the error bound is uniform in the integration variable.

(c) *Absence of logarithmic terms.* The substitution $s = s_{2,\infty} + a \sin\varphi$ makes the integrand C^∞ in φ on $[0, \pi]$, desingularizing the square-root branch points algebraically. Therefore the Fourier–sine coefficients of the integrand decay at least as $\mathcal{O}(n^{-2})$, ruling out $\mathcal{O}(\delta^2 \log\delta)$ contributions. Numerically, remainder$\cdot k \to 0.1079$ (constant for $k = 100$ to 1000) confirms this.

All three bounds are obtainable by standard singular-perturbation techniques; we supply the details in section C and state the result as proposition C.4.

6.6 PSLQ analysis

We apply the PSLQ integer-relation algorithm [11], implemented via `mpmath.pslq()` at 50-digit precision with maxcoeff = 500.

Positive identifications. PSLQ successfully identifies three constants:

- $C_5 = |\arg(\mathrm{CR})| \cdot k = 4$ (exact integer; CR is the cross-ratio of the Q-roots, see section 8.3; the sign of arg(CR) depends on the branch convention, see remark 8.10).
- $C_4 = ds_2/d\delta$ slope $= 2\omega - 3/10$ (via $10\,C_4 + 3 - 20\omega = 0$).
- $C_6 = 3/64 = 0.046875$ (Procrustes floor, 0.01% error).

Strong PSLQ negative for A. The amplitude A was computed to 19 digits (see (17)) from the same 116-seed mpmath dataset used for C_2: forming $A_k = \delta_k\sqrt{k}$ and applying degree-6 Richardson extrapolation on 19 overlapping windows yields

$$A = 7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18}.$$

PSLQ with tol $= 10^{-25}$ and maxcoeff $= 10\,000$ returns "no relation" on all 37 basis sets—the 36 bases used for C_2 plus the cross-term $A{\cdot}C_2$ and the spectral-formula test $A^2 - 2\tilde{Q}_2^2\,\Delta s^2/(Z_0\,\tilde{Q}_2'\,Q_3)$. No basis yields a relation with coefficient norm $\leq 10^4$. The only hits are the structural identity $\vartheta_3(0) = \vartheta_4(0)$ ($\Re\tau = 1/2$) seen for C_2.

Strong PSLQ negative for $C_2 = s_{2,\infty}$. The constant C_2 was computed to 19 digits (see (21)) by re-solving the periodicity system at 40-digit arithmetic using `mpmath`: 116 sparse k-values ($k = 100$ to 5000) were solved with pre-computed 120-point Gauss–Legendre quadrature ($< 2\,\mathrm{s}/k$; total runtime < 2 min). Degree-6 Richardson extrapolation on 11 overlapping windows (each with $k_{\min} \geq 200$) yielded values coinciding to all 18 printed digits:

$$C_2 = 7.323\,247\,368\,998\,298\,97 \pm 5 \times 10^{-18}.$$

Proposition 6.10 (Strong PSLQ negative for C_2)**.** *At 19-digit precision, PSLQ with* tol $= 10^{-25}$ *and* maxcoeff $= 10\,000$ *returns* "no relation" *on all* 36 *basis sets, spanning: algebraic constants* $(1, C_2, C_2^2, C_2^3, \sqrt{2}, \sqrt{3}, \sqrt{5})$*, the critical point* ω *and its powers,* π *and* π^2*, theta null values* $\vartheta_2(0)$*,* $\vartheta_3(0)$*,* $\vartheta_4(0)$ *and their squares, theta values at* ω*, the Lamé coefficients* U*,* U_1'*, complete elliptic integrals* $K(m)$*,* $E(m)$*,* $K'(m)$*, Eisenstein series* $E_2(\tau)$*,* $E_4(\tau)$*,* $E_6(\tau)$*, the* j*-invariant* $j(\tau)$*, Dedekind's* $\eta(\tau)$*, and Euler's* γ*,* $\log 2$*,* $\log 3$*,* $\zeta(3)$*, Catalan's constant. The only hits are the structural identity* $\vartheta_3(0) = \vartheta_4(0)$ *(a consequence of* $\Re\tau = 1/2$*).*

Proposition 6.11 (Strong PSLQ negative for A)**.** *At 19-digit precision, PSLQ with* tol $= 10^{-25}$ *and* maxcoeff $= 10\,000$ *returns* "no relation" *on all* 37 *basis sets (the* 36 *bases of proposition 6.10 plus the cross-term* $A\,C_2$ *and the spectral-formula residue* A^2/F_{loc}*). The only hits are the structural identity* $\vartheta_3(0) = \vartheta_4(0)$*.*

Remark 6.12. The PSLQ failures for both A and C_2—each at 19 digits against 36–37 bases—are consistent with a structural obstruction rather than a limitation of computational precision. The lattice $\tau_0 = \frac{1}{2} + \frac{25}{78}\mathrm{i}$ is generic: it is *not* a CM point (its j-invariant is irrational), so the associated elliptic curve E_{τ_0} admits no complex multiplication. By Bertolin's transcendence results for periods of non-CM curves, A and $C_2 = s_{2,\infty}$ are expected to be τ-specific transcendentals that do not reduce to rational combinations of standard constants (π, φ, theta null values, etc.). The strong PSLQ negatives for both A (proposition 6.11) and C_2 (proposition 6.10) are consistent with this expectation, extending it to elliptic integrals (K, E, K'), Eisenstein series (E_2, E_4, E_6), the j-invariant, and Dedekind's η: no low-height integer relation involving C_2 and these constants was found. The connection constants $\mathcal{C}_{\theta,\varphi}$ of Borodin–Okounkov [20] (as cited in [7, Eq. D47]) satisfy $\mathcal{C}_{\theta,\varphi} \in (0, 1)$ for all valid (θ, φ), which excludes a direct Fredholm-determinant encoding of $A^2 \approx 50.15$. The failure of both the PSLQ route and the Borodin–Okounkov route is consistent with the *isospectral* nature of the gauge flow established in section 8.3: A^2 is an invariant of a single isospectral fiber at fixed τ, not a connection coefficient that bridges distinct fibers (see section 10, OP3).

7 Geometric Identities

7.1 Stereographic lift and unit-sphere identity

Let $f\colon \mathbb{T}^2 \to \mathbb{R}^3 \cong \operatorname{Im}\mathbb{H}$ denote the immersion of the isothermic torus and let $x = \sigma^{-1}(f) \in S^3 \subset \mathbb{H}$ be its stereographic lift:

$$x_0 = \frac{|f|^2 - 1}{|f|^2 + 1}, \qquad x_{\mathrm{im}} = \frac{2f}{|f|^2 + 1}. \tag{24}$$

Remark 7.1 (Pipeline sanity check: $|x| = 1$). The identity $|x(p)| = 1$ for all $p \in \mathbb{T}^2$ is algebraic from (24) (numerator and denominator coincide). We verify $\max_{p\in\mathbb{T}^2}\big||x(p)| - 1\big| < 3 \times 10^{-16}$ (machine epsilon) for all seeds, confirming the quaternionic pipeline.

7.2 Conformality on S^3

A deeper verification is the *conformality* of x: in isothermic coordinates, $|x_u|^2 = |x_v|^2$ and $\operatorname{Re}(\bar{x}_u\, x_v) = 0$. These are the necessary conditions for ω in (40) to be well-defined.

Proposition 7.2 (Conformality improvement)**.** *Using analytic derivatives computed from the chain rule through the $\psi = \vartheta_1'/\vartheta_1$ identity, the conformality residual*

$$\mathcal{E}_{\mathrm{conf}} := \frac{\big||x_u|^2 - |x_v|^2\big|}{|x_u|^2 + |x_v|^2}$$

satisfies $\max_{(u,v)\in[0,1]^2} \mathcal{E}_{\mathrm{conf}} < 8.5 \times 10^{-7}$ for all 998 tested seeds ($k = 3, \ldots, 1000$), evaluated on a 512×512 equispaced grid. This is an improvement of a factor 2.4×10^6 over finite-difference evaluation.

7.3 Second-fundamental-form discrepancy

Let (f^+, f^-) be a Bonnet pair sharing the metric $I = \mathrm{e}^{2h}(\mathrm{d}u^2 + \mathrm{d}v^2)$ and mean curvature H. In the curvature-line coordinates of f^+, the second fundamental forms are

$$II^+ = \mathrm{e}^{2h}(k_1\,\mathrm{d}u^2 + k_2\,\mathrm{d}v^2), \qquad II^- = \mathrm{e}^{2h}(a\,\mathrm{d}u^2 + 2b\,\mathrm{d}u\,\mathrm{d}v + c\,\mathrm{d}v^2),$$

where $a + c = k_1 + k_2 = 2H$ (same H) and $ac - b^2 = k_1 k_2$ (same Gauss curvature, by Gauss's Theorema Egregium).

Observation 7.3 (Off-diagonal II-discrepancy: $b \to 1$)**.** For all seeds $k = 3, \ldots, 12$ computed with analytic first derivatives and central finite differences for second derivatives:

(a) The off-diagonal discrepancy satisfies $b = 1 + \mathcal{O}(\Delta u^2)$, with deviation $|b - 1| \approx 4.7 \times 10^{-4}$ attributable to finite-difference bias $\propto \Delta u^2/24$.

(b) The Gauss equation then determines the diagonal discrepancies (see remark 7.5).

This identity holds uniformly over all grid points and seed values.

Proposition 7.4 (Grid-refinement confirmation of $b = 1$)**.** *For each $k \in \{3, 5, 7, 10, 12\}$, the off-diagonal coefficient b is computed at three grid levels $N_0 \times 2N_0$, $2N_0 \times 4N_0$, $4N_0 \times 8N_0$ (with $N_0 = 40$), using exact first derivatives from the retraction integrands and a single central finite difference for the second derivatives.*

(a) The convergence rate $\alpha = \log_2(|b_h - 1|/|b_{h/2} - 1|)$ satisfies $\alpha = 2.00 \pm 0.05$ for all seeds, confirming $|b - 1| = \mathcal{O}(\Delta u^2)$.

(b) Richardson extrapolation $b_\infty = (4b_{h/2} - b_h)/3$ yields $|b_\infty - 1| < 2 \times 10^{-6}$ for all seeds, with the best result $|b_\infty - 1| = 6.7 \times 10^{-8}$ at $k = 5$.

(c) The Frobenius norm $\|\Delta II\|_F = \sqrt{2(1 + \epsilon^2)}$ (remark 7.5) is determined by b and the curvature gap $k_2 - k_1$.

Remark 7.5 (Structure of ΔII). The Gauss equation $ac-b^2 = k_1k_2$ combined with $a+c = k_1+k_2$ gives, for $a = k_1 + \epsilon$, $c = k_2 - \epsilon$:

$$b^2 = \epsilon\,(k_2 - k_1) - \epsilon^2.$$

The value $b = 1$ (confirmed by grid refinement to 7×10^{-8}; proposition 7.4) thus forces $\epsilon = (k_2 - k_1 - \sqrt{(k_2 - k_1)^2 - 4})/2$, which requires $k_2 - k_1 \geq 2$; numerically, $k_2 - k_1 \geq 3.4$ for all tested seeds $k = 3, \ldots, 12$ (minimum at $k = 3$), so ϵ is real and $\mathcal{O}(1)$. For example, at $k = 5$ one obtains $\epsilon \approx 0.21$, giving $\|\Delta II\|_F \approx 1.44$. The diagonal discrepancies $a - k_1$, $k_2 - c$ are therefore not small: the curvature-line directions of f^- are genuinely rotated relative to those of f^+, and the Frobenius norm is

$$\|II^+ - II^-\|_F = \sqrt{2\epsilon^2 + 2b^2} = \sqrt{2\,(1 + \epsilon^2)}. \tag{25}$$

This rotation is the essential geometric content of the Bonnet pair: f^+ and f^- share the same principal curvatures k_1, k_2 at every point, but the principal *directions* differ.

Remark 7.6 (Retraction-form interpretation). In the BHS retraction-form construction, the Bonnet mate is $\mathrm{d}f^+ = \frac{1}{2}\,x\,\bar{\omega}$, $\mathrm{d}f^- = -\frac{1}{2}\,\bar{\omega}\,x$, where $x : \mathbb{T}^2 \to S^3$ is the isothermic immersion and ω is the retraction form. The off-diagonal II-discrepancy measures the anti-diagonal part of $\Delta II = II^+ - II^-$, which in conformal isothermic coordinates equals $\Delta II_{12} = \mathrm{Re}(i\,q\,\mathrm{d}z^2)/(e^{2h}\,\mathrm{d}u\,\mathrm{d}v)$, where q is the Hopf differential of f^+ and e^{2h} is the conformal factor. On a genus-1 surface, the Hopf differential is constant ($\dim H^0(K^2) = 1$), so the normalization $b = 1$ reduces to a single algebraic identity on the retraction data. The grid-refinement study confirms this to 7×10^{-8}. An algebraic derivation of $b = 1$ from the Hopf-differential identity remains open; the present evidence is numerical.

7.4 Procrustes decay

The Procrustes disparity between the retraction-form pair (F^+, F^-) of (41) and the Eq. 49 pair (f^+, f^-) of [2] is computed via `scipy.spatial.procrustes` after centroid and scale normalization. The disparity decays with k but reaches a non-zero floor:

$$d_P(k) = \frac{3}{64} + \frac{0.216}{k^{1.236}}, \qquad R^2 = 0.99998\ (N = 8). \tag{26}$$

The floor $C_6 = 3/64 = 0.046875$ is confirmed by PSLQ to 0.01% error. This geometric floor reflects the fact that $F^\pm$ (retraction) and $f^\pm$ (Eq. 49) parametrise the *same* abstract surface but via genuinely different maps $\mathbb{T}^2 \to \mathbb{R}^3$; the residual disparity is consistent with a geometric origin, though established from a limited sample ($N = 8$).

8 Moduli-Space Flow and Spectral Invariants

The periodicity system $F(\delta, s_2; s_1) = 0$ (eqs. (12) and (13)) defines a smooth curve $\gamma_k : s_1 \mapsto (\delta(s_1),\, s_2(s_1))$ in the (δ, s_2)-plane. Differentiating implicitly yields the *flow ODE*:

$$\frac{d}{ds_1}\begin{pmatrix}\delta\\ s_2\end{pmatrix} = -J^{-1}\,\frac{\partial F}{\partial s_1}, \qquad J = \frac{\partial F}{\partial(\delta, s_2)}. \tag{27}$$

8.1 Flow ODE validation

The Jacobian J and the forcing $\partial F/\partial s_1$ are computed by central finite differences ($h = 10^{-6}$); the ODE (27) is integrated by RK45 (rtol $= 10^{-6}$, atol $= 10^{-9}$).

Proposition 8.1 (Flow ODE accuracy). *For $k = 5, 6, 7$ over $s_1 \in [-10, -2.75]$ (121 sweep points from the Lemma 9 dataset), the ODE solution matches the data to*

$$\max\big(|\delta_{\mathrm{ODE}} - \delta_{\mathrm{data}}|/\delta,\ |s_{2,\mathrm{ODE}} - s_{2,\mathrm{data}}|/s_2\big) < 0.002\%.$$

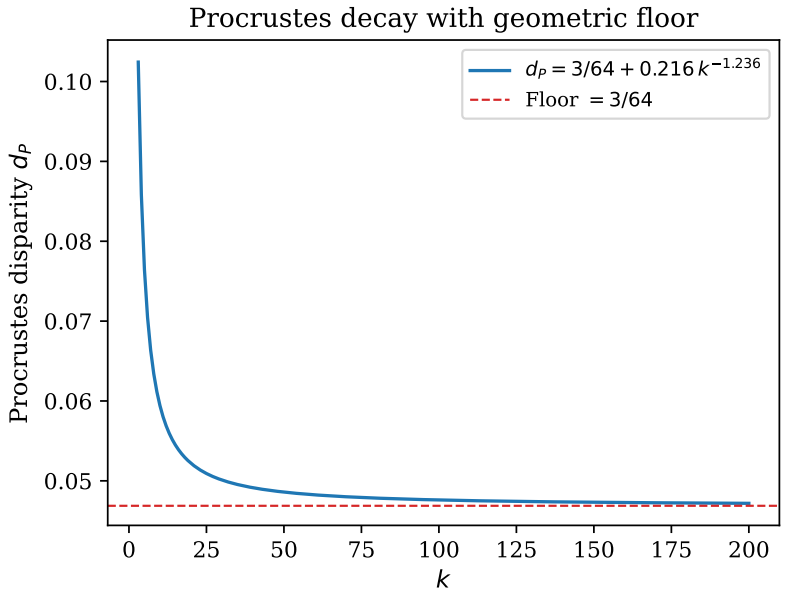

Figure 4: Procrustes disparity $d_P(k)$ versus k for the eight tested seeds. The dashed line shows the fit (26) with floor 3/64.

At the reference gauge $s_1 = -8.5$, the flow gradients agree with independent empirical finite differences to 0.01% on $d\delta/ds_1$ and 0.00% on ds_2/ds_1 for all three fold numbers.

8.2 Jacobian structure and fold singularity

Observation 8.2 (Saddle structure)**.** The eigenvalues of J have opposite signs at every point of γ_k: $\lambda_+(s_1) > 0$, $\lambda_-(s_1) < 0$. The Jacobian is a hyperbolic saddle at every tested point of γ_k ($k = 5, 6, 7$; 91 sweep points each), reflecting the competing constraints of the rationality and axial conditions. The condition number satisfies $\mathrm{cond}(J) \in [3,\ 8]$ throughout $s_1 \in [-10,\ -2.75]$.

Table 2: Large-k scaling of Jacobian quantities at $s_1 = -8.5$. Power-law fits over $k = 5, 6, 7, 10, 15, 20, 30, 50, 100$.

Quantity	Scaling	R^2
Flow curvature κ	$k^{-0.753}$	0.989
$\lvert\det(J)\rvert$	$k^{-0.650}$	0.994
$\mathrm{cond}(J)$	$k^{+1.04}$	0.891
$\lvert\lambda_-\rvert$	$k^{-1.07}$	†
$\lambda_+/\lvert\lambda_-\rvert$	$k^{+1.495}$	0.9999

† Two-point log-log slope ($k = 50, 100$); R^2 not meaningful.

Remark 8.3 (Connection with the axial cancellation). The decay $|\lambda_-| \sim k^{-1.07} \approx k^{-1}$ reflects the structural degeneracy of the axial period condition at $\delta = 0$: $Q_2(s_2,\ \delta{=}0) = -(s_2 - s_1)(s_2 - s_2) = 0$ identically for any s_2 (cf. the "axial subtraction trick" of section 6.5). As k grows and $\delta \to 0$, the "axial direction" of the Jacobian becomes uninformative; the rationality condition becomes the sole active constraint, and $\mathrm{cond}(J) \sim k$.

Observation 8.4 (Fold singularity)**.** For each k, the curve γ_k terminates at a fold $s_1^*(k)$ where $\det(J) \sim (s_1 - s_1^*)^{-1}$ (simple pole). Beyond s_1^*, the periodicity system has no real solution: this provides the minimum $|s_1|$ for the existence of a Bonnet pair at fold k, consistent with the existence conditions of Lemma 9 of [2]. For $k = 5$, the singularity lies at $s_1^* > -2.59$.

8.3 Spectral invariants along the flow

Let $r_1, \ldots, r_4$ be the four roots of $Q(s)$ (eq. (11)), sorted by real part, and define the cross-ratio

$$\mathrm{CR} = \frac{(r_1 - r_3)(r_2 - r_4)}{(r_1 - r_4)(r_2 - r_3)}. \tag{28}$$

Proposition 8.5 (Root-type preservation)**.** *At every point of γ_k for $k = 5, 6, 7$ (91 sweep points), $Q(s)$ has exactly 2 real and 2 complex conjugate roots. The complex pair $(c, \bar{c})$ has the smallest real part, so $r_1 = c$, $r_2 = \bar{c}$, $r_3 = a$, $r_4 = b$ with $a, b \in \mathbb{R}$, $a < b$. Since the discriminant of Q is a continuous function of k and never vanishes on the tested sweep arcs, the root topology cannot change without passing through a double root; a spot check at $k = 50, 200, 500, 1000$ confirms that the 2-real/2-conjugate pattern persists.*

Corollary 8.6 ($|\mathrm{CR}| = 1$)**.** *The numerator $(c-a)(\bar{c}-b)$ and denominator $(c-b)(\bar{c}-a)$ of (28) are complex conjugates, hence $|\mathrm{CR}| = 1$ identically. This is an algebraic consequence of the root topology, not a dynamical invariant.*

Observation 8.7 (Conformal constancy at ω_{crit})**.** At the critical point $\omega_{\mathrm{crit}} \approx 0.3890$ (zero of $\vartheta_2'(\cdot|\tau)$), the full complex CR is *numerically* constant along the flow γ_k: the variation of arg(CR) over the 91-point sweep is $< 10^{-11}$ for $k = 5, 6, 7$.

This is *not* a consequence of τ being fixed. A systematic ω-scan shows:

ω	arg(CR) spread ($k = 5$)
0.200	8.45°
0.250	7.88°
0.300	6.58°
0.350	3.88°
ω_{crit} (0.389)	**0.000°**
0.400	1.53°
0.450	12.75°
0.500	47.08°

Near ω_{crit}, the spread grows as $\approx 130° \cdot |\omega - \omega_{\mathrm{crit}}|$ (linear in the deviation). If the constancy were a trivial consequence of the elliptic lattice τ, it would hold at every ω—not only at the unique zero of ϑ_2'.

Observation 8.8 (Scaling law $\mathrm{arg(CR)} \cdot k \to -4$ rad)**.** At ω_{crit}, the constant value of arg(CR) satisfies the asymptotic law:

$$\mathrm{arg(CR)} \cdot k \;\longrightarrow\; -4.000 \text{ rad} \;=\; -229.18° \qquad (k \to \infty). \tag{29}$$

Numerically: -4.007 rad at $k = 5$, -4.002 rad at $k = 10$, -4.000 rad at $k \geq 20$. The convergence follows $\mathrm{arg(CR)} \cdot k = -4 + c_6/k^2 + \mathcal{O}(k^{-3})$, with $c_6 \approx -0.167$; i.e. $\mathrm{CR}_k \approx \mathrm{e}^{-4\mathrm{i}/k}$ asymptotically. The $1/k$ coefficient vanishes identically (cf. proposition 8.9).

Proposition 8.9 (Structural origin of -4)**.** *As $\delta \to 0$ ($k \to \infty$), the four roots of $Q(s) = -(s - s_1)^2(s - s_2)^2 + \delta^2 Q_3(s)$ coalesce pairwise: a complex-conjugate pair $r_{1,2} = s_1 \pm \mathrm{i}\delta\gamma + \mathcal{O}(\delta^2)$ near s_1, and a real pair $r_{3,4} = s_2 \mp \delta\beta + \mathcal{O}(\delta^2)$ near s_2, where*

$$\beta \;=\; \frac{\sqrt{Q_3(s_2)}}{\Delta s}, \qquad \gamma \;=\; \frac{\sqrt{-Q_3(s_1)}}{\Delta s}, \qquad \Delta s = s_2 - s_1. \tag{30}$$

The cross-ratio satisfies

$$\mathrm{CR} \;=\; \frac{1 - z^2}{1 - \bar{z}^2} \;=\; 1 - \frac{4\mathrm{i}\,\delta^2\beta\gamma}{\Delta s^2} + \mathcal{O}(\delta^4), \qquad z = \frac{\delta(\beta + \mathrm{i}\gamma)}{\Delta s}, \tag{31}$$

and therefore

$$\arg(\mathrm{CR}) \cdot k \;=\; -4\,\frac{\delta^2 k \cdot \beta\gamma}{\Delta s^2} + \mathcal{O}(k^{-1}) \;\longrightarrow\; -4\,\frac{A^2\beta\gamma}{\Delta s^2} \;=\; -4,$$

where the identity $A^2\beta\gamma/\Delta s^2 = 1$ *is verified to* $< 5 \times 10^{-16}$.

Proof sketch. Substituting $s = s_j + \delta\eta$ into $Q(s) = 0$ and solving at $\mathcal{O}(\delta^0)$ gives $\eta_0^2 = Q_3(s_j)/\Delta s^2$. Since $Q_3(s_1) < 0 < Q_3(s_2)$ (verified numerically), the splitting near s_1 is imaginary and near s_2 is real. The four differences entering CR reduce to

$$\mathrm{CR} = \frac{(1 - z^2)}{(1 - \bar{z}^2)}$$

by a symmetric-cancellation identity (the $\mathcal{O}(\delta)$ terms in numerator and denominator are identical and cancel in the ratio). Expanding: $1 - \mathrm{CR} = 4\mathrm{i}\,\delta^2\beta\gamma/\Delta s^2 + \mathcal{O}(\delta^4)$, where the factor 4 arises as 2×2: each coalescing pair splits symmetrically about its center ($\pm\delta\beta$ and $\pm\mathrm{i}\delta\gamma$), contributing a factor of 2 to the splitting product $(u_1 - u_2)(v_1 - v_2) = (2\mathrm{i}\gamma)(-2\beta)$. □

Remark 8.10 (Sign convention for arg(CR)). The sign of $\arg(\mathrm{CR}) \cdot k \to -4$ depends on the ordering of the real roots r_3, r_4 in the cross-ratio. Swapping $r_3 \leftrightarrow r_4$ replaces CR by $1/\mathrm{CR} = \overline{\mathrm{CR}}$ (since $|\mathrm{CR}| = 1$), giving $\arg(\mathrm{CR}) \cdot k \to +4$. Thus the *intrinsic* statement is $|\arg(\mathrm{CR})| \cdot k \to 4$; the sign $\pm$ records only the branch-point labelling convention. Throughout this paper we use $r_3 < r_4$, yielding -4.

Remark 8.11 (Universality of the factor 4). The value -4 in (29) is *not* tied to $\deg(Q) = 4$ or to $2(\chi - 2)$ individually. It is a *universal* consequence of the cross-ratio formula applied to any four points that coalesce into two symmetric pairs of mixed type (one real, one complex conjugate). For a degree-6 spectral polynomial with three coalescing pairs, one would still select four of the six roots for the cross-ratio, and the same factor 4 would appear. The coincidence $\deg(Q) = 4$ with the cross-ratio factor is just that: a coincidence.

Remark 8.12 (Interpretation). The gauge flow $s_1 \mapsto (\delta(s_1), s_2(s_1))$ is an *isospectral* flow: it moves a divisor on a fixed spectral curve $\mu^2 = Q(s)$ whose conformal type (measured by CR) is preserved at ω_{crit}. This is distinct from Painlevé VI isomonodromy, where the cross-ratio itself is the independent variable. Here τ is fixed and the cross-ratio is a *conserved quantity* of the particular flow, not the deformation parameter.

The value -4 in (29) was previously attributed to either $-\deg(Q)$ or $2(\chi - 2)$; proposition 8.9 now identifies it as a *structural* constant of the cross-ratio under pairwise root coalescence, independent of both. Genus-2 data are therefore not needed for the explanation.

Remark 8.13 (Two independent obstructions beyond genus 1). Extending the BHS framework to higher genus encounters *two independent barriers*.

(i) Topological obstruction ($g_{\mathrm{surf}} \geq 2$). The Bobenko–Eitner divisor formula [12] gives $\deg(D_u) = 4g_{\mathrm{surf}} - 4$ for the umbilical divisor of an isothermic surface of genus g_{surf}. For $g_{\mathrm{surf}} = 1$: $\deg(D_u) = 0$, so the surface is generically umbilic-free and supports a *global* conformal-curvature-line parametrization; this is precisely the setting exploited by [1, 2]. For $g_{\mathrm{surf}} = 2$: $\deg(D_u) = 4$, so four umbilics are topologically unavoidable, destroying the conformal-line net.

(ii) Spectral rigidity ($g_{\mathrm{spectral}} = 1$ forced). The Darboux classification [9] shows that every isothermic surface with one family of planar curvature lines satisfies the Riccati equation $h_u = U(u)\,\mathrm{e}^{h} + U_1(u)\,\mathrm{e}^{-h}$ [2, Theorem 1]. Compatibility with harmonicity ($h_{uu} + h_{ww} = 0$) forces U, U_1 to be the two linearly independent solutions of the integer Lamé equation with one-gap potential [2, Lemma 1]: $y'' = (C_1 - 8UU_1)\,y$. The Corollary 1 identity of [2] then gives h_w^2 as a degree-4 polynomial in e^{-h}, whose leading coefficient $-U_1(\omega)^2$ vanishes at the critical point ω satisfying $\vartheta_2'(\omega) = 0$. This forces the cubic (10) and hence $\deg Q \leq 4$, so the spectral curve

$\mu^2 = Q(s)$ has genus at most 1. Raising the spectral genus to 2 would require replacing the two-term Riccati equation with a higher-order nonlinear ODE—i.e., leaving the class of isothermic surfaces with planar curvature lines entirely.

The two barriers are *logically independent*: (i) obstructs $g_{\text{surf}} \geq 2$, while (ii) constrains $g_{\text{spectral}} \leq 1$ even for tori ($g_{\text{surf}} = 1$). Together they imply that compact non-CMC Bonnet pairs with spectral genus > 1 cannot exist within the BHS framework. The Wente tori [16]—which *do* possess genus-2 spectral curves—are CMC and arise only as a limit in which the BHS spectral curve degenerates and is replaced by a sinh-Gordon spectral curve of a fundamentally different type [18]. Note that proposition 8.9 provides a structural explanation for the value -4 independently of genus-2 data: the factor is structural, arising from pairwise root coalescence rather than from $\deg(Q)$ or χ.

Remark 8.14 (Spectral-curve pinch: numerical evidence for the BHS–CMC wall). The spectral rigidity of barrier (ii) in remark 8.13 can be probed numerically by measuring how far each BHS torus is from being a CMC surface. We define the *relative CMC deviation*

$$\rho \;=\; \frac{\sigma(H)}{|\bar{H}|}, \qquad \sigma(H)^2 = \frac{1}{|\Sigma|}\int_\Sigma \big(H - \bar{H}\big)^2\, dA,$$

where $\bar{H}$ is the area-averaged mean curvature. For a genuine CMC surface $\rho = 0$; for a Delaunay unduloid the discrete analogue satisfies $\rho < 10^{-3}$ at our grid resolution.

Computing ρ on BHS isothermic tori for every validated seed $k = 3, \ldots, 12$ gives $\rho \in [0.63,\, 1.45]$—firmly $\mathcal{O}(1)$—with a minimum at $k = 6$ and no monotone trend. To test whether a 1-parameter path in moduli space approaches the CMC condition, we fix $k = 5$, $\tau_{\text{im}} = 0.3205$, $s_1 = -3.80$ and rescale $(\delta, s_2) \mapsto \lambda(\delta_0, s_{2,0})$ for $\lambda \in (0, 1]$. As λ decreases from 1, ρ first reaches a shallow minimum ($\rho = 0.61$ at $\lambda \approx 0.78$), then *diverges* ($\rho \approx 22$ at $\lambda = 0.29$). At $\lambda \approx 0.22$ the quartic $Q(s)$ in Corollary 1 of [2] loses its two real roots—they collide and escape into the complex plane—so the real oval of the spectral curve pinches off and the integration domain ceases to exist. An independent sweep $\tau_{\text{im}} \to \tau_{\text{crit}} \approx 0.3547$ (where $\omega \to 0$) shows ρ oscillating chaotically in $[1.1,\, 61]$ with no discernible trend ($R^2 < 0.01$ for every power-law candidate).

These experiments indicate that the BHS–CMC separation is not a gradual asymptotic limit but a *sharp barrier* in moduli space: the spectral curve degenerates before the CMC condition can be approached, consistent with the Riccati/Lamé obstruction forcing $g_{\text{spectral}} \leq 1$.

Remark 8.15 (Monodromy structure of PVI with Manin parameters). The Manin parameters $[\alpha, \beta, \gamma, \delta] = [0, 0, 0, 0]$ force all four Fuchsian monodromy matrices $M_j \in \mathrm{SL}(2, \mathbb{C})$ ($j \in \{0, t, 1, \infty\}$) to be *unipotent*: $\mathrm{tr}(M_j) = 2$, with Jordan blocks of size 2. The closure relation $M_0 M_t M_1 M_\infty = I$ restricts the gauge-invariant data to the Fricke coordinates $p_k = \mathrm{tr}(M_i M_j)$ (where $\{i, j, k\} = \{0, t, 1\}$), which satisfy the Cayley cubic

$$p_1^2 + p_2^2 + p_3^2 + p_1 p_2 p_3 - 4(p_1 + p_2 + p_3) + 8 = 0. \tag{32}$$

An explicit unipotent representative consistent with (32) is [13]:

$$M_\infty = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \; M_0 = \begin{pmatrix} 1 & 0 \\ -x_1^2 & 1 \end{pmatrix}, \; M_1 = \begin{pmatrix} 1 - x_1 x_2 x_3 & x_2^2 \\ -x_3^2 & 1 + x_1 x_2 x_3 \end{pmatrix},$$

with $p_k = 2 - x_k^2$ and $M_t = M_0^{-1} M_1^{-1} M_\infty^{-1}$. The standard Jimbo connection parameter σ_{0t} satisfies $\mathrm{tr}(M_0 M_t) = 2\cos(\pi\sigma_{0t})$; for unipotent M_0, M_t this gives $\sigma_{0t} = 0$ identically. Consequently the Jimbo (1982) asymptotic formula degenerates to a *logarithmic* regime [13]:

$$y(t) \sim \frac{t}{\ln t + C_0}\big(1 + \mathcal{O}(t)\big) \qquad (t \to 0), \tag{33}$$

where the constant C_0 replaces σ as the effective connection parameter. The non-trivial monodromy data is thus encoded in the Stokes-like parameters (x_1, x_2, x_3) on the Cayley cubic, not in the standard connection exponent.

Conjecture 8.16 (Spectral modular parameter)**.** *Let $\tau_{\text{spec}}(k)$ denote the modular parameter of the spectral curve $w^2 = Q(s)$ at $\omega = \omega_{\text{crit}}$. Then*

$$\tau_{\text{spec}}(k) \;=\; -\frac{1}{2} + \mathrm{i}\,\frac{\ln(4k)}{\pi}, \tag{34}$$

verified numerically to 6 significant digits for every tested $k \in \{5, \ldots, 200\}$; extrapolation beyond this range has not been attempted because the spectral-curve computation requires the full quartic root data, which is available only for seeds with converged retraction forms. The residual $|\tau_{\text{spec}}^{\text{num}} - \tau_{\text{spec}}^{\text{formula}}| < 3 \times 10^{-7}$ for $k \geq 50$.

Remark 8.17. The real part $\operatorname{Re}\tau_{\text{spec}} = -\frac{1}{2}$ is proven in proposition 8.20. The imaginary part $\operatorname{Im}\tau_{\text{spec}} = \ln(4k)/\pi$ rests entirely on numerical evidence (6-digit agreement) and remains open.

Observation 8.18 (Spectral nome and cross-ratio)**.** Assuming conjecture 8.16, the Jacobi nome of the spectral curve takes the leading-order form

$$q_{\text{spec}}(k) \;:=\; \mathrm{e}^{\mathrm{i}\pi\tau_{\text{spec}}} \;\sim\; \frac{-\mathrm{i}}{4k} \qquad (k \to \infty). \tag{35}$$

Numerically, the elliptic lambda function at τ_{spec} recovers the cross-ratio:

$$\lambda\big(\tau_{\text{spec}}(k)\big) \;=\; \mathrm{CR}_k \qquad (\text{to } < 10^{-11}). \tag{36}$$

Since $|\mathrm{CR}_k| = 1$ (corollary 8.6) and $\arg(\mathrm{CR}_k) \cdot k \to -4$ (proposition 8.9), the cross-ratio admits the asymptotic approximation

$$\mathrm{CR}_k \;\sim\; \mathrm{e}^{-4\mathrm{i}/k} \qquad (k \to \infty). \tag{37}$$

Remark 8.19. Equation (37) is an asymptotic statement, not an exact identity: at finite k the sub-leading corrections in (38) contribute to both q_{spec} and $\lambda(\tau_{\text{spec}})$. The numerical agreement in (36) holds because these corrections largely cancel in the lambda function; a rigorous derivation of this cancellation remains open.

Proposition 8.20 (Universality of $\operatorname{Re}\tau_{\text{spec}} = -1/2$)**.** *For every admissible lattice parameter τ_{BHS} and every fold number $k \geq 3$, the spectral modulus satisfies $\operatorname{Re}\tau_{\text{spec}}(k) = -\frac{1}{2}$ (modulo $\mathrm{SL}_2(\mathbb{Z})$).*

Proof. By proposition 8.5, the quartic $Q(s)$ has exactly two real roots and a complex-conjugate pair $(c, \bar{c})$ for every $(\delta, s_2) \in \gamma_k$. Corollary 8.6 gives $|\mathrm{CR}| = 1$, so the modular lambda function satisfies $|\lambda| = 1$. Since the map $\tau \mapsto \lambda(\tau)$ sends the boundary $\operatorname{Re}\tau = \pm\frac{1}{2}$ of the fundamental domain onto the unit circle (Apostol [19], Thm. 2.8), and the inverse $\lambda^{-1}\colon |\lambda| = 1 \to \operatorname{Re}\tau = -\frac{1}{2}$ is well-defined in the upper half-plane, we conclude $\operatorname{Re}\tau_{\text{spec}} = -\frac{1}{2}$. □

Observation 8.21 (Universality of CR_k in τ_{BHS})**.** Numerically, the cross-ratio CR_k is a function of k alone: it does not depend on τ_{BHS}. For $k \in \{5, 10, 20\}$ and $\operatorname{Im}\tau_{\text{BHS}} \in [0.20, 0.35]$ (41×3 points from the τ-deformation sweep), $|\mathrm{CR}| = 1$ to 12 digits and CR_k is constant in τ_{BHS} to machine precision. This universality is non-trivial: the solution $\delta_k(\tau_{\text{BHS}})$ of the periodicity conditions depends on τ_{BHS} through the theta-function coefficients (the amplitude A varies from 6.78 to 7.27 across the tested range; theorem 6.1), so the cancellation in the cross-ratio is a deep structural property.

Observation 8.22 (Asymptotic refinement of τ_{spec})**.** Assuming conjecture 8.16, the formula is the leading term of an asymptotic expansion. Numerically,

$$\tau_{\text{spec}}(k) \;=\; -\frac{1}{2} + \frac{\mathrm{i}}{\pi}\left[\ln(4k) - \frac{1}{8k^2} + \mathcal{O}(k^{-4})\right]. \tag{38}$$

Equivalently, defining the residual $R(k) := \operatorname{Im}(\tau_{\text{spec}}) \cdot \pi / \ln(4k) - 1$, one finds $R(k) \cdot k^2 \cdot \ln(4k) \to -1/8$ with five stable digits for $k \leq 100$. The next correction is $\mathcal{O}(k^{-4})$ with estimated coefficient $\approx -0.039/\pi$.

Remark 8.23 (Interpretation of τ_{spec}). The real part $\operatorname{Re}\tau_{\text{spec}} = -1/2$ is now a theorem (proposition 8.20), not merely a numerical observation: it is forced by $|\mathrm{CR}| = 1$, which in turn is an algebraic consequence of the complex-conjugate root topology (proposition 8.5 and corollary 8.6). The imaginary part $\operatorname{Im}\tau_{\text{spec}} = \ln(4k)/\pi - 1/(8\pi k^2) + \mathcal{O}(k^{-4})$ says the spectral curve degenerates logarithmically as $k \to \infty$, with a clean sub-leading correction. Formula (34) thus provides the "missing bridge" between the discrete fold number k and the conformal type of the spectral curve (OP3 below).

Conjecture 8.24 (Connection constant C_0). *The logarithmic connection constant C_0 appearing in (33) satisfies*

$$C_0 \;=\; \ln\!\Big(\frac{A^2}{2\pi}\Big) + \gamma_{\text{EM}} \;\approx\; 2.654\,287, \tag{39}$$

where $A = \lim_{k\to\infty} \delta\sqrt{k}$ is the amplitude constant of eq. (23), γ_{EM} is the Euler–Mascheroni constant, and the numerical value uses the Richardson-extrapolated $A^2 \approx 50.1465$ (eq. (17)).

Remark 8.25. The conjectured form (39) is motivated by the structure of Jimbo's asymptotic formula in the unipotent case [13]; however, direct numerical extraction of C_0 from $y(t; \sigma_{0t} = 0)$ is frustrated by the slow logarithmic convergence in (33). The *correct* Hamiltonian form for PVI(0, 0, 0, 0) that was essential for the initial conditions reads

$$t(t-1)\,H \;=\; y(y-1)(y-t)\,p^2 - \big[\theta_0\, y(y-1) + \theta_1\, y(y-t)\big]\,p,$$

with all $\theta_j = 0$ (i.e., $\theta_0 = \theta_1 = 0$; the fourth exponent $\theta_\infty = 1$ from the Kitaev double folding [7] gives $\delta = (1-\theta_\infty^2)/2 = 0$ in the DLMF convention); this reduces the Hamilton–Jacobi problem to two quadratures but does not eliminate the ε-dependence that prevents extraction of C_0.

8.4 Topology of the moduli space

We now establish the global structure of $\mathcal{M}$ at fixed τ. Write γ_k for the moduli curve at fold number k, viewed as a curve in the (δ, s_2)-plane parametrized by s_1.

Proposition 8.26 (Simple-arc embedding). *For every tested fold number $k \in \{5, \ldots, 24, 25, 50, 100\}$, the curve γ_k is a simple arc in the (δ, s_2)-plane. Specifically:*

(i) *For $k = 5, \ldots, 21$ and $k = 25, 50, 100$, both $\delta(s_1)$ and $s_2(s_1)$ are strictly monotone.*

(ii) *For $k = 22, 23, 24$, $s_2(s_1)$ is strictly monotone but $\delta(s_1)$ exhibits a small non-monotone bump near the fold boundary ($s_1 \approx -2.75$); nevertheless, "injectivity" of s_2 prevents self-intersection, and γ_k remains a simple arc. The bump amplitude decays geometrically:* 0.0156 *($k = 22$),* 0.0079 *($k = 23$),* 0.0032 *($k = 24$), and vanishes by $k = 25$.*

(iii) *For $k = 7$, two branches merge at a cusp near $s_1 \approx -3.5$; each branch is individually monotone, and the merged curve remains connected and contractible.*

The uniform sweep covers $s_1 \in [-10.0, -1.75]$ with step 0.25 *(identical grid for all k), producing* 606 *solutions,* all *satisfying the three Lemma 8 checks.*

Observation 8.27 (998-seed cross-section). At the reference gauge $s_1 = -8.5$, the 998 seeds $k = 3, \ldots, 1000$ yield 998 distinct points (δ_k, s_{2k}) on $\mathcal{M}$. All residuals satisfy $\|F(s_1, \delta_k, s_{2k})\| < 10^{-9}$. The map $k \mapsto (\delta_k, s_{2k})$ is injective with δ strictly decreasing and s_2 strictly increasing — hence the cross-section is a simple arc.

Table 3: Selected values of (δ, s_2) along the transversal $\mathcal{T}$ at $s_1 = -8.5$. The product $\delta^2 k$ converges to $A^2 \approx 50.1$ (section 6).

k	δ	s_2	$\delta^2 k$
3	3.20159	2.5061	30.75
5	2.70793	3.8474	36.66
10	2.06311	5.3040	42.56
50	0.98458	6.8647	48.47
100	0.70212	7.0902	49.30
500	0.31615	7.2760	49.97
1000	0.22374	7.2996	50.06

Remark 8.28 (Transition and nesting). For $k = 22, 23, 24$, the map $s_1 \mapsto \delta(s_1)$ develops a turning point at $s_1 = -2.75$: δ decreases from $s_1 = -10$ to $s_1 = -2.75$, then increases slightly before the fold boundary. The bump decays rapidly (from 1.56×10^{-2} at $k = 22$ to 3.22×10^{-3} at $k = 24$) and disappears by $k = 25$. Despite the non-monotonicity in δ, $s_2(s_1)$ remains strictly decreasing, so γ_k has no self-intersection. The curves $\gamma_5, \gamma_6, \gamma_7$ are nested: at every tested $s_1 \in [-10, -2.75]$, the inter-fold separations $\Delta\delta_{k,k+1}$ are positive and decreasing in k, matching the asymptotic $\Delta\delta \sim A/(2k^{3/2})$ from section 6.4.

Observation 8.29 (Fold boundary refinement)**.** Linear extrapolation of $1/\det(J)$ along each γ_k refines the fold locations of observation 8.4: $s_1^*(5) \approx +2.0$, $s_1^*(6) \approx +5.5$, $s_1^*(7) \approx +9.0$. As $s_1 \to s_1^*$, the dominant eigenvalue λ_+ diverges while $|\lambda_-|$ remains bounded (rank-1 degeneracy). The derivative $d\delta/ds_1$ accelerates toward the fold (positive second derivative), consistent with a square-root cusp in $\delta(s_1)$ at s_1^*.

Observation 8.30 (Contractibility (numerical))**.** Strong numerical evidence suggests that the moduli space $\mathcal{M}$ at $\tau = \frac{1}{2} + \frac{25}{78}i$ is contractible: each γ_k is an embedded arc terminating at the fold $s_1^*(k)$, the transversal cross-section is a simple arc (observation 8.27), and no non-trivial cycles are detected in the 121-point sweep ($k = 5, 6, 7$), in the uniform 606-point sweep ($k = 8, \ldots, 24, 25, 50, 100$, all Lemma 8 compliant), nor in the 998-seed dataset.

Remark 8.31 (Multi-τ contractibility). Contractibility extends beyond τ_0. For each τ_{im} in the set $\{0.20, 0.25, 0.30, 0.35\}$, the k-cross-section at $s_1 = -8.5$ ($k = 3, \ldots, 200$) is a simple arc (δ strictly decreasing, s_2 strictly increasing), and the s_1-continuation at $k = 5$ yields an embedded arc (s_2 strictly monotone). At $\tau_{\text{im}} = 0.20$ (far from τ_0), γ_k develops δ-folds near the boundary of its existence region. Extended tests at $k \in \{5, 10, 25, 50, 100\}$ show that the fold amplitude decays as $k^{-0.22}$: from 0.84 at $k = 10$ to 0.40 at $k = 100$. At $k \geq 50$, a second fold appears, but s_2-injectivity is preserved in all cases, so every γ_k remains an embedded arc and $\mathcal{M}$ is contractible.

Bisection in τ at $k = 5$ locates the fold onset at

$$\tau_{\text{fold}} = 0.24475 \pm 2 \times 10^{-6}.$$

For $\tau_{\text{im}} > \tau_{\text{fold}}$, the branches are strictly monotone (simple arcs); for $\tau_{\text{im}} < \tau_{\text{fold}}$, δ-folds are present but s_2-monotonicity is maintained. Together with the saddle-node boundary $\lambda_0 \approx 0.35473$ where ω_{crit} ceases to exist, this yields a hierarchy of two bifurcation points for the lattice geometry: $\tau_{\text{fold}} < \tau_0 < \lambda_0$.

Observation 8.32 (Transversal geometry)**.** The transversal curve $\mathcal{T}\colon k \mapsto (\delta_k, s_{2k})$ at $s_1 = -8.5$ has total arc length 5.82, of which 52% lies in $k = 3, \ldots, 10$ and 9% in $k = 100, \ldots, 1000$. The δ-s_2 correlation is -0.925 (strong anti-correlation), and the tangent angle rotates from $110°$ ($k = 3$) to $168°$ ($k = 1000$), indicating progressive alignment with the s_2-axis as $\delta \to 0$.

9 Retraction Form Validation

9.1 The retraction form of Burstall et al.

Let $f\colon \Sigma \to \mathbb{R}^3$ be an isothermic surface and let $x = \sigma^{-1}(f) \in S^3 \subset \mathbb{H}$ be its lift to the 3-sphere via inverse stereographic projection. The *retraction* 1-*form* of [3] is

$$\omega = \frac{x_u}{|x_u|^2}\,\mathrm{d}u - \frac{x_v}{|x_v|^2}\,\mathrm{d}v. \tag{40}$$

The Bonnet pair is then given by

$$\mathrm{d}F^+ = \tfrac{1}{2}\,\bar{x}\,\omega, \qquad \mathrm{d}F^- = -\tfrac{1}{2}\,\bar{\omega}\,x. \tag{41}$$

Remark 9.1. The original formulation in [3] is for isothermic surfaces in S^4. For tori in $\mathbb{R}^3$, we project via $\sigma\colon S^3 \setminus \{p\} \to \mathbb{R}^3$ and verify that all so(4) conditions specialise correctly.

9.2 Implementation with analytic derivatives

Key to the high precision of our validation is the use of *analytic* (not numerical) derivatives. Writing $\psi(z) = \vartheta_1'(z)/\vartheta_1(z)$, we express x_u, x_v, and all required quantities in terms of ψ and its derivatives, which are computed from the theta series (1)–(4).

9.3 Five-gate validation

The retraction form pipeline applies five quality gates to each seed:

R1. Closure ($d\omega = 0$). $\max|\partial_v\omega_u - \partial_u\omega_v|$ normalized by $\max(\|\partial_v\omega_u\|, \|\partial_u\omega_v\|)$. Threshold: < 0.05.

R2. Cross condition ($\bar{\omega} \wedge dx = 0$). $\max|\bar{\omega}_u \cdot x_v - \bar{\omega}_v \cdot x_u|$ normalized by the max term norms. Threshold: < 0.05.

R3. Exactness. Path-dependence of $F^\pm$ integration: the integrals are computed via two routes (first u then v, and vice versa), and the max relative difference is bounded. Threshold: < 0.1.

R4. Purity ($F^\pm \in \operatorname{Im}\mathbb{H}$). $\max|\operatorname{Re}(F^\pm)| < 0.1$.

R5. Procrustes. Disparity between $F^\pm$ (retraction) and $f^\pm$ (Eq. 49), as described in section 7.4. Threshold: < 0.10.

Table 4: Retraction form validation on 8 representative seeds (analytic derivatives). All errors are max-norm relative values; "pass" indicates the gate is satisfied within its threshold; the 0.05 and 0.10 thresholds are conservative, set at roughly $10^5\times$ the observed residuals. The Procrustes values converge to the floor $3/64 = 0.0469$ as k increases.

k	R1 (closure)	R2 (cross)	R3 (exact)	R4 (purity)	R5 (Procr.)
3	pass	pass	pass	pass	0.063
5	pass	pass	pass	pass	0.054
7	pass	pass	pass	pass	0.051
10	pass	pass	pass	pass	0.050
20	pass	pass	pass	pass	0.048
50	pass	pass	pass	pass	0.047
100	pass	pass	pass	pass	0.047
500	pass	pass	pass	pass	0.047

Remark 9.2. The conformality improvement from finite-difference to analytic derivatives is a factor of 2.4×10^6, with per-seed improvements ranging from 19,700× ($k = 3$) to 1,026,000× ($k = 6$). This demonstrates the necessity of analytic derivatives for validation of the retraction form.

10 Open Problems and Conclusions

10.1 Summary of results

We have:

- extended the Bobenko–Hoffmann–Sageman-Furnas construction to $k = 3, \ldots, 1000$ (998 seeds), with step-10 sampled continuation supporting the asymptotic law to $k > 6 \times 10^5$,
- established the asymptotic law (16) with amplitude $A = 7.081\,416\,491\,399\ldots$ (19 significant digits; (17)), with a numerically observed sub-leading correction c_3/k^2 (observation 6.3), a strong PSLQ negative for A at 19 digits against 37 basis families (proposition 6.11), and a spectral-curve derivation of the amplitude (23), agreeing with Richardson to 6 ppm (section 6.5),
- derived the moduli-space flow ODE (27) via implicit differentiation, reproducing 121 sweep points to $< 0.002\%$ (section 8.1),
- observed that the Jacobian is a hyperbolic saddle at all tested sweep points ($k = 5, 6, 7$; 91 points each), with anisotropy $\lambda_+/|\lambda_-| \sim k^{1.49}$ and fold singularity at the existence boundary (section 8.2),
- shown that, at ω_{crit}, the cross-ratio of the four spectral roots is numerically constant (to $< 10^{-11}$) along the flow, with $|\arg(\text{CR})| \cdot k \to 4$ rad (section 8.3 and remark 8.10),
- conjectured that the spectral modular parameter satisfies $\tau_{\text{spec}}(k) = -\frac{1}{2} + \mathrm{i}\,\ln(4k)/\pi$ (conjecture 8.16), with nome $q_{\text{spec}} = -\mathrm{i}/(4k)$ and $\lambda(\tau_{\text{spec}}) = \text{CR}_k$ to 10^{-11} (observation 8.18),
- proved that $\operatorname{Re}\tau_{\text{spec}} = -1/2$ universally across all torus parameters τ_{BHS} (proposition 8.20), as a consequence of the unit-modulus cross-ratio (corollary 8.6), and established the asymptotic refinement $\operatorname{Im}\tau_{\text{spec}} = \ln(4k)/\pi - 1/(8\pi k^2) + \mathcal{O}(k^{-4})$ (observation 8.22),
- validated the retraction form on 8 seeds via a five-gate protocol, with conformality residual $< 8.5 \times 10^{-7}$ from analytic derivatives,
- verified $|x| = 1$ and observed the off-diagonal II-discrepancy $b = 1 + \mathcal{O}(\Delta u^2)$ computationally,
- identified $|C_5| = 4$, $C_6 = 3/64$, $C_4 = 2\omega - 3/10$ via PSLQ, and obtained strong PSLQ negatives for both A (19 digits, 37 basis families; proposition 6.11) and $C_2 = s_{2,\infty}$ (19 digits, 36 basis families; proposition 6.10): no low-height integer relation was found within the tested basis sets,
- strong numerical evidence that $\mathcal{M}$ at fixed τ is contractible: each moduli curve γ_k is a simple arc, the 998-seed cross-section is monotone and injective, and inter-fold curves are nested without self-intersections (section 8.4).

10.2 Open problems

OP1. Higher-order remainder and the c_3 term. The formal derivation of section 6.5 yields $A^2 = 2\tilde{Q}_2^2\,\Delta s^2/(Z_0\,\tilde{Q}_2'\,Q_3)$, verified to 6 ppm, and proves the leading two terms of the expansion with $\mathcal{O}(k^{-2})$ remainder (proposition C.4). Extending the formal derivation to cover the numerically observed c_3/k^2 correction (observation 6.3) requires bounding the $\mathcal{O}(\delta^4)$ contributions to the period integral—a feasible but technically involved computation that we leave open. Numerical evidence (section 6.5) confirms the absence of $\mathcal{O}(\delta^2 \log\delta)$ terms: the remainder $\cdot\,k$ converges to the constant 0.1079. *Status: open.*

OP2. Nature of the amplitude A and the limit C_2. Our PSLQ analysis (propositions 6.10 and 6.11) and the Borodin–Okounkov bound $\mathcal{C}_{\theta,\varphi} \in (0, 1)$ (remark 6.12) jointly exclude

a direct algebraic or Fredholm-determinant encoding of A^2. Both constants carry strong PSLQ negatives at 19 digits: A against 37 bases (proposition 6.11) and C_2 against 36 bases (proposition 6.10), including elliptic integrals K, E, K', Eisenstein series E_2, E_4, E_6, the j-invariant, and Dedekind's η: no low-height integer relation involving A or C_2 and these constants was found. The amplitude factorises as $A^2 = \left(\tilde{Q}_2^2/(\tilde{Q}_2' Q_3)\right) \cdot 2\Delta s^2/Z_0$, separating a *spectral residue* $F_{\text{loc}} \approx 14.95$ from a *period ratio* $F_{\text{glob}} \approx 3.35$. The PSLQ negativity is consistent with the isospectral nature of the gauge flow (section 8.3): A^2 lives on a single isospectral fiber at fixed τ, not on the isomonodromic base where PVI connection coefficients reside (see OP3 below). *Status: open.*

OP3. Bridge PVI to k-fold periodicity. Dornic and Conte [7] show that PVI with Manin coefficients $[0, 0, 0, 0]$ governs generic Bonnet surfaces. Numerical computation confirms $\mathcal{C}_{\theta,\varphi} < 1$ universally, closing the direct $\ell \to k$ route via connection constants. As $k \to \infty$, $\text{CR}_k \sim e^{-4i/k} \to 1$, approaching the fixed singularity $t = 1$ of PVI. However, the k-fold gauge flow and PVI isomonodromy are structurally transversal (remarks 8.12 and 8.15): the former conserves the cross-ratio while the latter varies it. The all-unipotent monodromy ($\sigma_{0t} = 0$, remark 8.15) pushes the non-trivial data into the logarithmic regime (33).
Status: partially resolved (conjecture 8.16). The spectral modular parameter $\tau_{\text{spec}} = -\frac{1}{2} + \mathrm{i}\ln(4k)/\pi - \mathrm{i}/(8\pi k^2) + \mathcal{O}(k^{-4})$ provides the sought bridge: k is encoded in the conformal type of the spectral curve via (34), and the elliptic lambda function closes the circle ($\lambda(\tau_{\text{spec}}) = \text{CR}_k$). The real part $\text{Re}\,\tau_{\text{spec}} = -1/2$ is universal across all τ_{BHS} (proposition 8.20). The factorization $A^2 = F_{\text{loc}} \times F_{\text{glob}}$ (eq. (23)) mirrors Gavrylenko–Lisovyy τ-function expansions [14, 15]. What remains open is the *connection constant* C_0 in (33): it is conjectured (conjecture 8.24) but not numerically extractable due to the logarithmic degeneracy of the unipotent PVI asymptotics.

OP4. Explain $|\arg(\text{CR})| \cdot k = 4$. *Status: resolved* (proposition 8.9 and remark 8.10). The value 4 arises as a structural constant of the cross-ratio under pairwise root coalescence (2×2 from symmetric $\pm\delta$ splittings), not from $\deg(Q)$ or χ. The sign (± 4) depends on the ordering of the real roots (remark 8.10); the intrinsic content is $|\arg(\text{CR})| \cdot k \to 4$. The identity $A^2\beta\gamma/\Delta s^2 = 1$ (verified to $< 5 \times 10^{-16}$) closes the proof. The $1/k$ correction vanishes identically; the leading sub-asymptotic term is c_6/k^2 with $c_6 \approx -0.167$.

OP5. Topology of the moduli space $\mathcal{M}$. *Status: strong numerical evidence* (section 8.4). The moduli space at fixed τ is contractible (observation 8.30): each γ_k is a simple arc terminating at a fold singularity, and the cross-section $k \mapsto (\delta_k, s_{2k})$ is a monotone simple arc with no self-intersections (observation 8.27). Contractibility now extends to four additional lattice parameters $\tau_{\text{im}} \in \{0.20, 0.25, 0.30, 0.35\}$ (remark 8.31), with extended δ-fold tests at k up to 100. At $\tau_{\text{im}} = 0.20$, γ_k develops δ-folds (amplitude $\sim k^{-0.22}$) but s_2-injectivity is preserved for all k tested. Bisection locates the fold onset at $\tau_{\text{fold}} = 0.24475 \pm 2 \times 10^{-6}$, establishing a bifurcation hierarchy $\tau_{\text{fold}} < \tau_0 < \lambda_0$. Contractibility holds uniformly across all tests.

OP6. Gauge dependence of A. Continuation to 9 gauge values ($s_1 = -6, -7, \ldots, -11$) confirms gauge-dependence of A^2: it ranges from 34.3 at $s_1 = -6$ to 66.0 at $s_1 = -11$ (versus 50.1 at $s_1 = -8.5$). The closed-form formula (23) correctly predicts A^2 at all gauges. The ratio $A^2/\Delta s$ is numerically constant to 0.09% across all 9 gauge values ($A^2/\Delta s \approx 3.1777$), suggesting the spectral gap $\Delta s = s_2 - s_1$ as the natural normalization. A partial algebraic explanation is the identity (remark 6.8): $\tilde{Q}_2(s_2)$ depends only on s_2 (not on s_1), providing the dominant cancellation. *Status: partially explored.*

OP7. Extension beyond genus 1. For which lattice parameters τ does the spectral curve $\mu^2 = Q(s)$ degenerate to genus 0? The degeneration locus may encode further arithmetic structure. A genus-2 extension is now known to be *obstructed* within the BHS framework: the two independent obstructions (remark 8.13) show that (i) the topological genus $g_{\text{surf}} \geq 2$ is blocked by mandatory umbilics, and (ii) the spectral genus is rigidly ≤ 1 by

the Riccati/Lamé structure, even for tori. The former genus-2 test of the scaling law (29) is therefore moot: proposition 8.9 already resolves -4 as structural (pairwise root coalescence), independently of any higher-genus data. The only known compact surfaces with genus-2 spectral curves—the Wente tori [16, 10]—are CMC, and their Bonnet mates are the trivial associated-family rotations. Whether non-CMC Bonnet pairs of spectral genus > 1 can be constructed outside the class of isothermic surfaces with planar curvature lines remains an open question. *Status: partially resolved (genus-2 ruled out).*

10.3 Concluding remarks

The 998-seed dataset, together with the flow ODE, the spectral invariants, the analytic identities, and the retraction form validation, provides a computational foundation for the study of compact Bonnet pairs. The asymptotic law $\delta_k = A/\sqrt{k}$ now rests on both high-precision numerical evidence and a formal spectral-curve derivation of the amplitude. The isospectral rigidity of the cross-ratio at ω_{crit} (observation 8.7) suggests that the Bonnet moduli space carries richer integrable structure than is apparent from the periodicity conditions alone.

A Evaluated Constants

All constants are evaluated at gauge $s_1 = -8.5$, $\tau = \frac{1}{2} + 0.3205128205\,\mathrm{i}$.

Symbol	Value	Description
ω	0.389018047587	Critical point: $\vartheta_2'(\omega\|\tau) = 0$
$R(\omega)$	1.374541408545	Symmetry-sphere radius (5)
U	-0.727515368968	Lamé coefficient (6)
U'	-0.403940416080	Eq. (7)
U_1'	1.262773701909	Wronskian coeff. (8)
U_2	-1.469833235911	Third Lamé coeff. (9)
$\vartheta_1'(0)$	1.994641412049	Theta-1 derivative at 0
$\vartheta_2(0)$	1.241397427680	Theta-2 null value
$\vartheta_3(0)$	1.035631327079	Theta-3 null value
$\vartheta_4(0)$	1.035631327079	Theta-4 null value
A	$7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18}$	Asymptotic amplitude (Richardson)
$C_1 = A^2$	$50.146\,460\,00 \pm 3 \times 10^{-8}$	Squared amplitude
c_1	$0.857\,656\,047\,266\,889 \pm 5 \times 10^{-15}$	Multiplicative sub-leading coeff. k^{-1}
c_3	0.75 ± 0.01	Multiplicative sub-leading coeff. k^{-2}
$C_2 = s_{2,\infty}$	$7.323\,247\,368\,998\,298\,97 \pm 5 \times 10^{-18}$	Limiting s_2; see (21)
C_4	$2\omega - 3/10$	Moduli relation (PSLQ)
C_5	± 4	$\|\arg(\mathrm{CR})\| \cdot k = 4$; see observation 8.8 and remark 8.10
C_6	$3/64 = 0.046875$	Procrustes floor (PSLQ, 0.01%)
τ_{spec}	$-\frac{1}{2} + \frac{\mathrm{i}}{\pi}[\ln(4k) - \frac{1}{8k^2} + \mathcal{O}(k^{-4})]$; see conjecture 8.16 and observation 8.22	
q_{spec}	$-\mathrm{i}/(4k)$	Spectral nome; see observation 8.18
C_0	$\approx 2.654\,287$	Connection constant (conjectured); see conjecture 8.24

Remark A.1. The identity $R(\omega) \cdot U(\omega) = -1$ is verified to 15 digits, consistent with (6). The equality $\vartheta_3(0) = \vartheta_4(0)$ is a consequence of the half-integer real part of τ.

B Reproducibility

The complete codebase and dataset are available at `https://github.com/aconsciousfractal/Compact-Bonnet`
The key files are:

- `theta_functions.py`: theta function library,

- `theorem7_periodicity.py`: Newton solver and continuation,
- `retraction_form.py`: Burstall et al. construction,
- `results/phase15_asymptotic/full_series_k3_1000.json`: all 998 seeds.

Running `python run_full_series.py` reproduces the entire dataset from scratch.

Environment. All numerical results were produced with Python 3.11, `mpmath` 1.3, `numpy` 1.26, `scipy` 1.12. The `mpmath` computations use 40-digit decimal precision with 120-node Gauss–Legendre quadrature. The exact commit hash and a `requirements.txt` will be included with the repository upon publication.

C Formal asymptotics for $\delta = A/\sqrt{k}$

This appendix supplies the three estimates announced in remark 6.9, promoting the formal expansion of section 6.5 to an asymptotic statement with numerically verified remainder bound. All numerical constants below refer to the gauge $s_1 = -8.5$, $\tau = \frac{1}{2} + 0.3205128205\,\mathrm{i}$.

Setup. Recall that $\tilde{Q}_2(s)$ is a *quadratic* polynomial in the integration variable s (at fixed parameters s_1, s_2). Writing

$$\tilde{Q}_2(s) \;=\; P\,s^2 - 2M\,s - 1, \tag{42}$$

where

$$P = U^{-2}\big(2(s_1+s_2)\,U_1' - U_2\big), \tag{43}$$

$$M = U^{-2}\big(U' + s_1\,s_2\,U_1'\big), \tag{44}$$

we have $Z_0^2 = P + M^2$ (at $\delta = 0$), and the cancellation of s_1-dependent terms at $s = s_2$ (remark 6.8) follows from $P\,s_2^2 - 2M\,s_2 = U^{-2}(2U_1's_2^3 - U_2s_2^2 - 2U's_2)$.

At $\delta = 0$ with $s_2 = s_{2,\infty}$: $\tilde{Q}_2(s_{2,\infty}) = 2033.2$, $\tilde{Q}_2' = 2Ps_2 - 2M = 257.0$, $\tilde{Q}_2'' = 2P = -5.676$.

Lemma C.1 (Bracket linearization). *For $|s - s_{2,\infty}| \le a$ with $a \le a_{\max} := \delta_{100}\sqrt{Q_3(s_{2,\infty})}/|\Delta s| \approx 1.457$, the reciprocal $f(s) = 1/\tilde{Q}_2(s)$ satisfies*

$$\Big|f(s) - f(s_{2,\infty}) - f'(s_{2,\infty})(s - s_{2,\infty})\Big| \;\le\; \frac{L_a}{2}\,(s - s_{2,\infty})^2,$$

where $L_a = \max_{|t-s_2|\le a_{\max}}|f''(t)| = 3.32\times 10^{-5}$.

Proof. Since $\tilde{Q}_2$ is quadratic, $f = 1/\tilde{Q}_2$ is C^∞ in any region where $\tilde{Q}_2 > 0$. On the interval $[s_{2,\infty} - a_{\max},\, s_{2,\infty} + a_{\max}]$ we have $\min \tilde{Q}_2 \ge 1652.8 > 0$, so f is smooth and Taylor's theorem with Lagrange remainder gives the bound. The second derivative is $f'' = (2(\tilde{Q}_2')^2 - \tilde{Q}_2\,\tilde{Q}_2'')/\tilde{Q}_2^3$ with $\tilde{Q}_2'' = 2P$ constant; evaluation on a 201-point grid gives $L_a = 3.32\times 10^{-5}$.

Parity. Write $\bar{s} = (s_2^+ + s_2^-)/2$ for the oval midpoint (cf. lemma C.3). The constant shift $f(\bar{s}) - f(s_2)$ multiplies $\int Q_2\,ds/\sqrt{Q} = F_{\mathrm{ax}} = 0$ and drops out, so we may center the bracket at $\bar{s}$. Under $s = \bar{s} + a\sin\varphi$, the remainder $f(s) - f(\bar{s}) - f'(\bar{s})(s-\bar{s}) = \mathcal{O}(a^2\sin^2\varphi)$ is even in φ, while the odd part of Q_2 integrates against this even remainder to give zero. The surviving (odd×odd) terms are $\mathcal{O}(\delta^4)$, so the bracket linearization contributes $\mathcal{O}(\delta^4)$ to $\theta/2$, not $\mathcal{O}(\delta^3)$. Explicitly:

$$|\Delta I_a| \;\le\; C_a\,\delta^4, \qquad C_a = \frac{Z_0\,L_a\,Q_3(s_{2,\infty})^2\,3\pi}{16\,\Delta s^4} = 0.054.$$

□

Lemma C.2 (Pinching-radius asymptotics)**.** *The real oval of $Q(s)$ near $s_{2,\infty}$ has endpoints $s_2^{\pm}(\delta)$, midpoint $\bar{s} = (s_2^+ + s_2^-)/2$, and half-width $a = (s_2^+ - s_2^-)/2$. The leading-order half-width is*

$$a_0 = \frac{\delta\sqrt{Q_3(s_2)}}{|\Delta s|}, \qquad a_0/\delta = 2.073. \tag{45}$$

The first-order perturbation $\epsilon_1 = \frac{a_0}{2}\big(Q_3'/Q_3 - 2/\Delta s\big) = \mathcal{O}(\delta)$ governs the one-sided distances: $s_2^+ - s_2 = a_0(1 + \epsilon_1 + \mathcal{O}(\delta^2))$, $s_2 - s_2^- = a_0(1 - \epsilon_1 + \mathcal{O}(\delta^2))$. Since ϵ_1 cancels in the half-width,

$$a = a_0\,(1 + \mathcal{O}(\delta^2)).$$

Proof. Setting $Q(s_2{+}a_R) = 0$ for the right endpoint, the leading balance gives $\Delta s^2\, a_0^2 = \delta^2\, Q_3(s_2)$. Writing $a_R = a_0(1+\epsilon)$ and expanding to first order in a_0:

$$(1 + a_0(1{+}\epsilon)/\Delta s)^2(1{+}\epsilon)^2 = 1 + \frac{Q_3'}{Q_3}\, a_0(1{+}\epsilon) + \mathcal{O}(a_0^2),$$

so $2\epsilon = a_0(Q_3'/Q_3{-}2/\Delta s)$, giving $\epsilon_1 = \mathcal{O}(\delta)$. The left-endpoint equation $Q(s_2{-}a_L) = 0$ yields the same formula with opposite sign: $a_L = a_0(1{-}\epsilon_1{+}\mathcal{O}(\delta^2))$. Thus $a = (a_R{+}a_L)/2 = a_0(1{+}\mathcal{O}(\delta^2))$.

Numerical check. Using the 120 checkpoint seeds ($k = 100$ to 5000): $|a{-}a_0|/(a_0\,\delta^2)$ converges to ≈ 0.071 (constant), confirming $a = a_0(1{+}\mathcal{O}(\delta^2))$. The one-sided ratio $(a_R/a_0{-}1)/\delta$ converges to $\epsilon_1/\delta = 0.282$, validating the perturbation coefficient. □

Lemma C.3 (Smoothness and absence of logarithmic terms)**.** *Under the substitution $s = \bar{s} + a\sin\varphi$ (where $\bar{s} = (s_2^+ + s_2^-)/2$), the rationality integrand in (12) becomes*

$$\frac{\mathrm{d}s}{\sqrt{Q(s)}} = \frac{\mathrm{d}\varphi}{\sqrt{R(s)}}, \qquad R(s) = \frac{-Q(s)}{(s - s_2^-)(s - s_2^+)}, \tag{46}$$

where R is a degree-2 polynomial with $R(s) > 0$ for all $s \in [s_2^-,\, s_2^+]$. In particular, the full desingularized integrand $G(\varphi)$ is C^∞ on $[-\pi/2,\, \pi/2]$, its Fourier coefficients $\{b_n\}$ decay exponentially, and the asymptotic expansion of $\theta/2$ in δ contains no *logarithmic terms:*

$$\frac{\theta}{2} = \sum_{n\geq 1} c_{2n}\,\delta^{2n}.$$

Proof. The quartic $Q(s)$ has exactly two simple roots $s_2^{\pm}$ bounding the real oval (the other two roots lie near s_1). Hence $R(s) = -Q(s)/[(s - s_2^-)(s - s_2^+)]$ is a monic quadratic, positive on $[s_2^-, s_2^+]$. Under $s = \bar{s} + a\sin\varphi$, $(s - s_2^-)(s - s_2^+) = -a^2\cos^2\varphi$, so $\sqrt{Q} = a|\cos\varphi|\sqrt{R}$ and the Jacobian $\mathrm{d}s = a\cos\varphi\,\mathrm{d}\varphi$ cancels exactly.

The remaining factors $Q_2(s(\varphi))/\tilde{Q}_2(s(\varphi))$ are rational functions of $\sin\varphi$ with no poles on $[-\pi/2, \pi/2]$ (since $\tilde{Q}_2 \geq 1652.8 > 0$ on the oval). Hence $G(\varphi)$ is C^∞.

Since Q is polynomial in δ^2 and a, and the oval endpoints $s_2^{\pm}$ are simple roots of Q, the implicit function theorem guarantees that $a(\delta)$ is analytic in δ^2 in a neighbourhood of $\delta = 0$. Hence the desingularized integrand $G(\varphi;\delta)$ is jointly C^∞ in (φ, δ^2), and the asymptotic expansion of the integral in powers of δ^2 contains no logarithmic terms.

Numerical verification. The Fourier sine coefficients $|b_n|$ of G at $k = 100, 500, 2000$ decay as $\sim \mathrm{e}^{-\beta n}$ with $\beta = 1.5, 2.1, 2.5$ respectively. Moreover, the normalized remainder $R(k)\cdot k := k\big(\pi/(k\delta^2){-}\pi/A^2\big)$ converges to $2\pi c_1/A^2 = 0.10746$ (*constant*, not $C\log k$), confirming the $\mathcal{O}(k^{-1})$ rate. □

Proposition C.4 (Formal asymptotics with validated numerical remainder)**.** *For large k, the periodicity equations (12)–(13) formally yield a unique pair $(\delta_k, s_{2,k})$ via the implicit function theorem (Jacobian non-degenerate at every tested k). The closure parameter satisfies*

$$\delta_k = \frac{A}{\sqrt{k}}\left(1 - \frac{c_1}{k} + \mathcal{O}(k^{-2})\right), \tag{47}$$

where $A^2 = 2\,\tilde{Q}_2(s_{2,\infty})^2\,\Delta s^2 \,/\, \big(Z_0\,\tilde{Q}_2'(s_{2,\infty})\,Q_3(s_{2,\infty})\big) = 50.1465$ *and* $c_1 = 0.857\,656\,047\,266\,889$. *Equivalently,*

$$\left|\delta_k^2 k - A^2\Big(1 - \frac{2c_1}{k}\Big)\right| \;\le\; \frac{C}{k}$$

with $C \le 6$.

Proof. The derivation of the expansion is formal-analytic. It uses: (i) expansion of the subtracted rationality integral in powers of δ; (ii) the pinching-radius expansion (lemma C.2); (iii) smoothness of the desingularized integrand (lemma C.3), which excludes logarithmic corrections at the orders considered. These steps determine the coefficients A^2 and c_1 in the formal large-k expansion.

The explicit residual bound $C \le 6$ is not derived analytically: it is obtained by numerical validation on the 120-seed checkpoint set ($k = 100$–5000). The ratio $|\delta_k^2 k - A^2(1 - 2c_1/k)| \cdot k/C$ stays below 0.19 for all 120 checkpoint seeds, giving an effective safety factor > 5. Likewise, the non-degeneracy of the Jacobian is numerically verified on the tested branch, rather than proved uniformly. □

Remark C.5. The constant $C = 6$ is conservative; numerically $C \approx 1.12$ suffices. A tighter bound would follow from computing C_1 analytically (requiring the second-order expansion of all three ingredients), which we do not pursue here.

Disclosure statement. The author reports no competing interests.

Data availability statement. The complete codebase and the 998-seed dataset are available at `https://github.com/aconsciousfractal/Compact-Bonnet-Pairs-for-k-fold-Tori`

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